

Chapter 1 – Introduction

1.1 Background

The rapid growth of digital technologies, cloud computing, interconnected systems, Internet of Things (IoT) devices, and online services has significantly increased the dependence of individuals and organizations on computer networks and information systems. Along with these technological developments, the cybersecurity landscape has become increasingly complex. Cyber attackers continuously develop new techniques to exploit vulnerabilities, compromise systems, steal sensitive information, disrupt services, and conduct financial or politically motivated attacks. Malware, phishing, ransomware, distributed denial-of-service (DDoS) attacks, exploitation of vulnerabilities, and credential-based attacks are among the major threats faced by modern organizations.

The increasing volume, velocity, and complexity of cyber threats have made traditional security approaches insufficient for identifying and responding to threats in a timely manner. Security systems generate large quantities of logs, alerts, incident reports, vulnerability information, threat reports, and other forms of security-related data. A considerable portion of this information is unstructured or semi-structured textual data obtained from security blogs, incident reports, vulnerability databases, security advisories, online forums, news articles, and other sources. Manually examining such information requires significant time and specialized expertise and may result in delays or inconsistencies in threat analysis.

Cyber Threat Intelligence (CTI) has emerged as an important component of modern cybersecurity. CTI involves the collection, processing, analysis, and dissemination of information related to cyber threats and threat actors. The objective of CTI is not simply to collect security information but to transform available data into meaningful intelligence that can support security monitoring, threat detection, incident response, risk assessment, and decision-making.

However, the continuously increasing amount of CTI data presents a major analytical challenge. Cybersecurity analysts must identify relevant information, determine the type of threat, extract important indicators, understand relationships among events, and recognize changes in threat patterns. Automated techniques based on Machine Learning (ML), Natural Language Processing (NLP), and topic modeling can assist in processing large-scale textual CTI data and extracting useful patterns.

Threat classification is one important task in automated CTI analysis. Classification systems can categorize threat-related information into predefined classes such as malware, phishing, ransomware, DDoS, vulnerability exploitation, and other attack categories. Automated classification can reduce the effort required for manual analysis and enable security teams to organize and prioritize large amounts of threat information.

Another important challenge is the detection of emerging threats. Existing classification systems generally depend on predefined categories and labeled datasets. Consequently, they may have difficulty identifying new or evolving threats that do not fit existing categories. Emerging threats may initially appear as changes in the frequency, terminology, entities, techniques, or topics discussed across CTI sources. Temporal topic modeling provides a way to analyze such changes by examining how topics evolve over time.

Therefore, this research focuses on developing a framework for **real-time cyber threat classification and emerging threat detection using Machine Learning and temporal topic modeling**. The proposed research combines supervised threat classification with temporal analysis of cyber threat-related textual information. The objective is to support automated identification of known threat categories while also identifying changing or emerging threat patterns from CTI data.

1.2 Cyber Threat Intelligence

Cyber Threat Intelligence (CTI) is the process of collecting, processing, analyzing, and communicating information about cyber threats in a form that can support cybersecurity decisions and actions. CTI enables organizations to move beyond the simple detection of individual security events by providing contextual information about threats, threat actors, attack techniques, vulnerabilities, and potential impacts.

Modern CTI is generated from multiple sources, including security reports, vulnerability databases, malware analysis reports, security advisories, incident reports, online forums, blogs, news articles, and other publicly available or organizational sources. A significant amount of this information is textual and unstructured, making automated processing an important research area.

The CTI lifecycle generally involves several interconnected activities:

1. **Planning and Direction** – defining intelligence requirements and determining what information is required.
2. **Collection** – gathering relevant threat information from appropriate sources.
3. **Processing** – cleaning, normalizing, structuring, and preparing collected information for analysis.
4. **Analysis** – extracting meaningful patterns, relationships, indicators, and threat information.
5. **Dissemination** – delivering analyzed intelligence to relevant stakeholders.
6. **Feedback** – evaluating the usefulness of intelligence and refining future collection and analysis activities.

Machine Learning and NLP techniques can support multiple stages of this lifecycle, particularly processing and analysis. Automated models can identify patterns in large datasets that may be difficult to discover through manual analysis.

1.2.1 Definition of Cyber Threat Intelligence

Cyber Threat Intelligence can be defined as evidence-based knowledge regarding existing or potential cyber threats that is collected, processed, analyzed, and communicated to support informed cybersecurity actions. Unlike raw security data, CTI provides context and interpretation that can help security professionals understand the nature, source, behavior, and potential consequences of cyber threats.

CTI may contain information about threat actors, malware families, attack techniques, vulnerabilities, indicators of compromise (IoCs), targeted organizations, affected technologies, and observed attack behaviors. The usefulness of CTI depends on the quality, timeliness, relevance, and accuracy of the information.

For automated cyber threat analysis, textual CTI reports are particularly valuable because they contain descriptions of attacks, vulnerabilities, techniques, tools, threat actors, and observed behaviors. Extracting useful information from these reports requires appropriate NLP and ML techniques.

1.2.2 Types of Cyber Threat Intelligence

Cyber Threat Intelligence is commonly categorized into four major types: strategic, tactical, operational, and technical intelligence.

Strategic Cyber Threat Intelligence

Strategic CTI provides high-level information intended primarily for decision-makers and organizational leadership. It focuses on broader trends, risks, threat landscapes, potential business impacts, and long-term cybersecurity considerations.

Strategic intelligence can assist organizations in understanding how changes in the cyber threat environment may affect business operations and security investments.

Tactical Cyber Threat Intelligence

Tactical CTI focuses on the techniques, tactics, and procedures used by threat actors. It helps security professionals understand how attackers operate and which defensive controls may be required.

Examples include information about attack techniques, exploitation methods, malware behavior, and attacker procedures.

Operational Cyber Threat Intelligence

Operational CTI provides information about ongoing or planned cyber campaigns and attacks. It may include information about threat actors, campaigns, targeted sectors, attack timelines, and observed activities.

Operational intelligence can support incident response and security operations by providing contextual information about active threats.

Technical Cyber Threat Intelligence

Technical CTI focuses on technical indicators associated with cyber attacks. Examples include IP addresses, domain names, URLs, file hashes, malware signatures, email addresses, and other Indicators of Compromise (IoCs).

Technical CTI is frequently used by security tools such as Security Information and Event Management (SIEM), intrusion detection systems, endpoint security platforms, and threat intelligence platforms.

The four types of CTI are complementary. Strategic intelligence supports long-term decisions, tactical intelligence describes attacker techniques, operational intelligence provides information about campaigns, and technical intelligence provides actionable technical indicators.

1.3 Problem Statement

The increasing volume of cyber threat information has created significant challenges for cybersecurity analysts and organizations. Large quantities of CTI are continuously generated from security reports, vulnerability disclosures, incident analyses, online sources, and other threat-related information channels. Much of this information is unstructured textual data, making manual analysis time-consuming and difficult to scale.

Existing approaches to cyber threat analysis face several challenges.

1.3.1 Large Volume of Unstructured CTI Data

Cyber threat information is generated continuously and often contains lengthy textual descriptions. Manually reading and analyzing large numbers of reports requires substantial time and human effort.

1.3.2 Difficulty in Automated Threat Classification

Threat reports may describe different attack types using varying terminology and writing styles. A classification system must identify meaningful textual patterns to accurately categorize threat information.

1.3.3 Dependence on Predefined Threat Categories

Many supervised classification approaches require predefined labels. Such systems are effective for known threat categories but may have difficulty handling previously unseen or evolving threat patterns.

1.3.4 Difficulty in Detecting Emerging Threats

New cyber threats can emerge gradually through changes in attack techniques, terminology, tools, vulnerabilities, and threat-actor behavior. Conventional classification systems may not explicitly model these temporal changes.

1.3.5 Temporal Changes in Cyber Threat Information

The characteristics of cyber threats change over time. A topic that is relatively uncommon in one period may become significantly more prominent later. Therefore, analyzing CTI without considering its temporal dimension may result in the loss of important information about emerging trends.

1.3.6 Need for Timely Analysis

Cybersecurity operations require timely identification of threats. Delayed processing of threat information can reduce the practical value of intelligence. Automated approaches are therefore required to process and analyze threat information efficiently.

Based on these challenges, there is a need for an automated framework that can classify known cyber threats while also analyzing temporal changes in CTI data to identify potentially emerging threat patterns.

1.4 Research Motivation

The primary motivation of this research is the increasing complexity and continuously changing nature of the cyber threat landscape. Cyber attacks evolve rapidly, and new vulnerabilities, malware families, attack techniques, and campaigns can emerge over time. Security teams therefore require automated methods that can process large quantities of threat intelligence and identify meaningful patterns efficiently.

Traditional manual analysis is difficult to scale because analysts must examine large volumes of heterogeneous information. Machine Learning provides an opportunity to automate the classification of threat-related information based on patterns learned from historical data.

However, classification alone may not be sufficient for identifying new or evolving threats. A threat classification model generally operates according to categories present in its training data. Emerging threats may initially appear as previously unseen combinations of terms, entities, techniques, or topics. This creates a need for complementary analytical techniques that can examine changes in threat-related information over time.

Temporal topic modeling can provide such complementary analysis by identifying topics within textual CTI data and examining how their distributions change across different time periods. A sudden increase or significant change in a topic may provide an indication of an emerging threat pattern that requires further investigation.

The combination of supervised classification and temporal topic analysis therefore provides the motivation for this research. The proposed approach aims to support both **known threat identification** and **emerging threat pattern analysis** within a unified CTI framework.

1.5 Research Objectives

The main objective of this research is to develop a framework for automated cyber threat classification and emerging threat detection using Machine Learning and temporal topic modeling.

The specific objectives are:

1. **To collect and prepare cyber threat intelligence data** from relevant threat reports and textual CTI sources.
 2. **To preprocess CTI textual data** using appropriate Natural Language Processing techniques such as text normalization, tokenization, stop-word removal, and feature preparation.
 3. **To develop Machine Learning-based classification models** for automatically categorizing cyber threat information into predefined threat classes.
 4. **To compare the performance of different classification algorithms** using appropriate evaluation metrics such as accuracy, precision, recall, and F1-score.
 5. **To apply topic modeling techniques** to identify important topics and thematic patterns within cyber threat intelligence data.
 6. **To incorporate temporal analysis into topic modeling** to investigate how threat-related topics change across different time periods.
 7. **To identify potentially emerging threat patterns** by analyzing changes in topic distributions, frequencies, and temporal trends.
 8. **To develop an integrated framework** combining threat classification and temporal topic analysis for automated CTI analysis.
 9. **To evaluate the effectiveness of the proposed framework** using appropriate experimental datasets and performance measures.
 10. **To analyze limitations and potential improvements** for future development of automated cyber threat intelligence systems.
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1.6 Research Scope

The scope of this research is focused on the automated analysis of textual Cyber Threat Intelligence using Machine Learning, Natural Language Processing, and temporal topic modeling.

The research includes the following areas:

- Collection and preparation of textual cyber threat intelligence data.
- Text preprocessing and normalization.
- Feature extraction and text representation.
- Machine Learning-based cyber threat classification.
- Comparative evaluation of classification algorithms.
- Topic modeling of CTI documents.
- Temporal analysis of extracted topics.
- Identification of changing and potentially emerging threat patterns.
- Evaluation using classification and topic-analysis metrics.
- Development of an integrated analytical framework for CTI.

The research primarily focuses on **textual CTI reports and threat-related documents**. The framework is intended to analyze information contained within the available dataset rather than replace operational cybersecurity systems.

The research does not attempt to provide a complete enterprise Security Operations Center (SOC), automated incident-response platform, or offensive cybersecurity system. It also does not claim that every newly detected temporal topic represents a confirmed zero-day vulnerability or previously unknown cyber attack. Emerging-topic detection is treated as an analytical mechanism for identifying patterns that may require further cybersecurity investigation.

The effectiveness of the proposed framework will depend on factors such as dataset quality, availability of sufficient labeled data, class distribution, language characteristics, temporal coverage, and the quality of extracted textual features.

1.7 Contributions

The major contributions of this research are as follows:

1.7.1 Automated Cyber Threat Classification

The research develops an automated Machine Learning-based approach for categorizing textual CTI into predefined cyber threat classes. This reduces the dependence on completely manual classification of large collections of threat reports.

1.7.2 Comparative Evaluation of Machine Learning Models

Multiple Machine Learning approaches are evaluated using consistent experimental conditions and standard classification metrics. The comparison provides an understanding of how different models perform on the selected CTI dataset.

1.7.3 Temporal Topic Analysis of CTI

The research incorporates temporal analysis into topic modeling to investigate how cyber threat topics evolve across different time periods. This provides an additional perspective beyond conventional document-level classification.

1.7.4 Emerging Threat Pattern Identification

The proposed framework analyzes changes in topic distributions and temporal trends to identify potentially emerging threat patterns. Such patterns can be used as signals for further cybersecurity analysis.

1.7.5 Integrated CTI Analysis Framework

The research combines supervised threat classification with temporal topic modeling into an integrated analytical framework. This enables the analysis of both known threat categories and changing threat-related topics.

1.7.6 Experimental Analysis

The proposed approach is evaluated using quantitative metrics and comparative experiments. The analysis includes classification performance, confusion matrices, precision, recall, F1-score, topic distributions, temporal trends, and emerging-threat analysis.

1.7.7 Reproducible Research Workflow

The research follows a structured data-processing and experimentation workflow covering data preparation, model training, evaluation, topic modeling, temporal analysis, and visualization. This provides a reproducible foundation for further research in automated CTI analysis.

1.8 Thesis Organization

This thesis is organized into the following chapters:

Chapter 1 – Introduction

This chapter introduces the research background and provides an overview of Cyber Threat Intelligence. It discusses the problem statement, research motivation, objectives, scope, major contributions, and organization of the thesis.

Chapter 2 – Literature Review

This chapter presents a detailed review of existing research related to Cyber Threat Intelligence, Machine Learning-based cyber threat classification, Natural Language Processing, topic modeling, temporal

analysis, and emerging cyber threat detection. The chapter also identifies limitations and research gaps in existing approaches.

Chapter 3 – Research Design and Data Preparation

This chapter describes the research design, data sources, dataset characteristics, data collection process, dataset construction, preprocessing procedures, feature representation, and experimental data preparation.

Chapter 4 – Proposed Methodology

This chapter presents the proposed framework for real-time cyber threat classification and emerging threat detection. It describes the overall architecture, Machine Learning classification pipeline, topic modeling approach, temporal analysis, emerging-threat detection mechanism, and evaluation methodology.

Chapter 5 – Experimental Results and Analysis

This chapter presents the experimental setup and results obtained from the proposed methodology. It includes dataset distribution analysis, model implementation, classification performance, comparative model analysis, confusion matrices, precision, recall and F1-score, temporal topic modeling, emerging-threat detection, real-time detection analysis, error analysis, discussion, and limitations.

Chapter 6 – Conclusion and Future Work

This chapter summarizes the major findings and contributions of the research. It discusses the conclusions derived from the experimental results and presents potential directions for future research, including improved temporal modeling, larger CTI datasets, advanced language models, real-time data sources, and enhanced emerging-threat detection techniques.

Chapter 2 – Literature Review

2.1 Cyber Threat Intelligence

Cyber Threat Intelligence (CTI) has become an important component of modern cybersecurity because of the increasing complexity, frequency, and diversity of cyber attacks. CTI focuses on collecting, processing, analyzing, and disseminating information related to cyber threats so that organizations can make informed decisions regarding cybersecurity operations.

Traditional cybersecurity systems primarily focus on detecting individual security events, whereas CTI provides contextual information that can help explain the nature of a threat, the techniques used by an attacker, the targeted assets, and potential consequences. This contextual information can support security monitoring, incident response, vulnerability management, risk assessment, and strategic decision-making.

CTI is commonly divided into strategic, tactical, operational, and technical intelligence. Strategic intelligence provides high-level information about the broader threat landscape and potential organizational impact. Tactical intelligence focuses on attacker tactics, techniques, and procedures (TTPs). Operational intelligence provides information about ongoing or planned campaigns, while technical intelligence contains technical indicators such as IP addresses, domains, URLs, file hashes, and malware signatures.

The CTI lifecycle generally consists of planning and direction, collection, processing, analysis, dissemination, and feedback. Each stage contributes to converting raw cybersecurity information into actionable intelligence. However, the continuous growth of CTI sources has created challenges related to data volume, heterogeneity, timeliness, and analysis.

Recent research has increasingly investigated automated CTI analysis using Natural Language Processing (NLP), Machine Learning (ML), Deep Learning (DL), and knowledge-based approaches. These techniques can process large volumes of textual information and extract useful patterns that may be difficult to identify manually.

The base research underlying this thesis demonstrates the use of Machine Learning and topic modeling for automated cyber threat classification and emerging-threat analysis. The study highlights the potential of combining classification with topic-based analysis to support automated CTI processing.

2.2 CTI Data Sources

The effectiveness of an automated CTI system depends significantly on the quality and diversity of its data sources. Cyber threat information is distributed across multiple sources, including structured databases, security reports, vulnerability repositories, security blogs, online forums, incident reports, and threat intelligence platforms.

2.2.1 Security Reports

Security reports published by cybersecurity organizations often contain detailed descriptions of attacks, malware, vulnerabilities, threat actors, and observed attack techniques. These reports are particularly useful for NLP-based CTI research because they contain rich textual descriptions.

2.2.2 Vulnerability Databases

Vulnerability databases provide structured information about known security vulnerabilities. The Common Vulnerabilities and Exposures (CVE) system is an important example. Vulnerability records can contain descriptions, affected products, severity information, and references.

Such information can be used to investigate vulnerability trends and relationships between vulnerabilities and observed attack techniques.

2.2.3 Malware Reports

Malware analysis reports provide information about malicious software, including malware families, capabilities, infection mechanisms, indicators of compromise, and behaviors. These reports are valuable sources for studying malware-related threats.

2.2.4 Security Blogs and Advisories

Cybersecurity companies and research organizations regularly publish security blogs and advisories describing newly discovered threats, vulnerabilities, campaigns, and attack techniques. These sources can provide relatively recent information about changes in the threat landscape.

2.2.5 Online Forums and Underground Sources

Cybersecurity-related forums and underground communities can contain information about vulnerabilities, malicious tools, credentials, campaigns, and attacker discussions. However, such sources can contain noisy, incomplete, duplicated, or unreliable information and therefore require careful preprocessing and validation.

2.2.6 News and Incident Reports

Cybersecurity news and incident reports provide information about major attacks, breaches, ransomware campaigns, and security incidents. Their temporal characteristics make them useful for studying the emergence and evolution of cyber threats.

2.2.7 CTI Platforms and Structured Feeds

Threat intelligence platforms can provide structured indicators and relationships between threat entities. Examples include feeds containing domains, IP addresses, URLs, hashes, malware identifiers, and other indicators.

For this research, the primary focus is on **textual CTI reports and threat-related documents**, as textual sources provide sufficient information for threat classification and temporal topic analysis.

2.3 Cyber Threat Classification

Cyber threat classification is the process of assigning cyber threat information to predefined categories based on characteristics contained in the data. Classification can help security analysts organize large collections of threat reports and prioritize information for further investigation.

Threat categories may include malware, phishing, ransomware, denial-of-service attacks, vulnerability exploitation, credential attacks, botnets, and other forms of malicious activity.

Traditional threat classification relies heavily on manually defined rules, signatures, and expert analysis. Although such approaches remain useful for specific detection tasks, they can become difficult to maintain when threat descriptions and attack techniques continuously change.

Machine Learning provides an alternative by learning classification patterns from previously labeled examples. A typical ML-based CTI classification pipeline includes data collection, preprocessing, feature extraction, model training, prediction, and evaluation.

Text representation methods such as Bag-of-Words, Term Frequency-Inverse Document Frequency (TF-IDF), Word2Vec, and Doc2Vec have been investigated for converting threat-related text into numerical representations.

The quality of a classification system depends on several factors, including dataset size, label quality, class balance, text representation, model selection, and the similarity between training and unseen data.

A major limitation of supervised classification is its dependence on predefined classes. If a new type of threat does not correspond to an existing training category, a conventional classifier may incorrectly assign it to an existing class. This limitation motivates the use of complementary techniques such as topic modeling and temporal analysis.

2.4 NLP for CTI

Natural Language Processing (NLP) provides techniques for automatically processing and analyzing human language. Since a large portion of CTI is available as textual information, NLP has become an important technology for automated threat intelligence analysis.

A typical NLP pipeline for CTI may include:

1. Text cleaning
2. Tokenization
3. Lowercase conversion
4. Removal of punctuation and irrelevant characters
5. Stop-word removal
6. Stemming or lemmatization
7. Named entity extraction
8. Feature extraction
9. Text representation

Cybersecurity text presents several challenges that distinguish it from general-domain text. CTI documents frequently contain technical terminology, product names, malware families, vulnerability identifiers, IP addresses, domain names, file hashes, abbreviations, and specialized security terminology.

Consequently, conventional NLP preprocessing techniques must be carefully adapted. For example, indiscriminate removal of special characters may remove meaningful information such as CVE identifiers, IP addresses, URLs, or hashes.

2.4.1 Text Representation

Traditional NLP-based CTI systems commonly use sparse representations such as Bag-of-Words and TF-IDF.

The TF-IDF representation assigns a weight to a term based on its frequency within a document and its frequency across the complete collection. It can be represented as:

$$TFIDF(t,d) = TF(t,d) \times IDF(t)$$

where $TF(t,d)$ represents the frequency of term t in document d , and $IDF(t)$ represents the inverse document frequency of the term.

Word embedding approaches such as Word2Vec and Doc2Vec provide dense representations and can capture semantic relationships between words or documents. However, these representations may not fully capture context-dependent meanings.

The development of contextual language models has subsequently enabled more advanced representations of cybersecurity text.

2.5 Machine Learning for CTI

Machine Learning has been extensively investigated for automated cybersecurity analysis. In CTI, ML techniques can be applied to classification, clustering, anomaly detection, entity extraction, malware analysis, and threat prediction.

2.5.1 Supervised Learning

Supervised learning uses labeled examples to train a model to predict predefined categories. Common algorithms include:

- Naïve Bayes
- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree
- Random Forest
- Gradient Boosting
- k-Nearest Neighbors (k-NN)

These algorithms have been used for text classification because of their relatively straightforward implementation and computational efficiency.

SVM-based methods can be particularly effective for high-dimensional sparse text representations. Random Forest can capture nonlinear relationships and is less dependent on assumptions about the underlying feature distribution.

2.5.2 Unsupervised Learning

Unsupervised learning operates without predefined labels. It can be used to discover hidden structures and groups within CTI datasets.

Clustering algorithms such as K-Means and hierarchical clustering can group documents based on their feature representations. Such approaches can be useful when labeled CTI data are limited.

However, interpreting automatically generated clusters can be difficult because the resulting groups may not directly correspond to meaningful cybersecurity categories.

2.5.3 Evaluation of ML-Based CTI Classification

Common evaluation metrics include accuracy, precision, recall, and F1-score.

Accuracy is defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision is:

$$Precision = \frac{TP}{TP + FP}$$

Recall is:

$$Recall = \frac{TP}{TP + FN}$$

The F1-score combines precision and recall:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

For imbalanced CTI datasets, relying only on accuracy can be misleading. Therefore, precision, recall, F1-score, and class-wise performance should also be considered.

2.6 Deep Learning for CTI

Deep Learning extends traditional Machine Learning by using multi-layer neural networks to learn increasingly complex representations from data. Deep learning approaches have been applied to several cybersecurity tasks, including malware detection, intrusion detection, threat classification, and security-text analysis.

2.6.1 Convolutional Neural Networks

Convolutional Neural Networks (CNNs) can identify local patterns in sequential data and have been applied to text classification. In CTI analysis, CNN-based architectures can learn local word or character patterns that may be associated with specific threat categories.

2.6.2 Recurrent Neural Networks

Recurrent Neural Networks (RNNs) are designed to process sequential information. Variants such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) can capture dependencies across sequences and have been used in cybersecurity text classification and event analysis.

However, recurrent architectures process sequences sequentially, which can make training less efficient for very long documents.

2.6.3 Limitations of Traditional Deep Learning

Although CNN, RNN, LSTM, and GRU models can learn useful representations automatically, they may face limitations when processing long documents or modeling relationships between distant words.

These limitations contributed to the development and adoption of attention-based architectures and Transformer models.

2.7 Transformer Models

Transformer models have significantly changed the field of Natural Language Processing by introducing attention-based mechanisms for modeling relationships between tokens.

The Transformer architecture was introduced to address limitations associated with sequential processing in recurrent neural networks. Its self-attention mechanism allows the model to consider relationships between different parts of a sequence simultaneously.

A simplified scaled dot-product attention mechanism is expressed as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where Q , K , and V represent the query, key, and value matrices, respectively, and d_k represents the dimensionality of the key vectors.

Transformer-based models such as BERT and its variants have demonstrated strong performance in many NLP tasks. Domain-specific models have also been developed to better represent specialized language.

For cybersecurity applications, Transformer models can be used for:

- Threat report classification
- Malware-related text analysis
- Vulnerability information extraction
- Threat entity recognition
- Attack technique identification
- Security event classification
- CTI document representation

One important advantage of Transformer-based representations is their ability to capture contextual relationships between words. The meaning of a cybersecurity term can depend strongly on its surrounding context, and contextual representations can capture this information more effectively than traditional word representations.

However, Transformer models may require substantial computational resources and large datasets for effective training or fine-tuning. Their computational requirements can also make deployment challenging in resource-constrained environments.

For this research, Transformer-based representations can be considered as an advanced feature representation or comparative approach alongside conventional Machine Learning methods.

2.8 Topic Modeling

Topic modeling is an unsupervised learning technique used to discover latent thematic structures within a collection of documents. Unlike supervised classification, topic modeling does not require predefined document labels.

Topic modeling is particularly relevant to CTI because threat reports may contain information about previously unknown or changing threat categories. By identifying groups of frequently co-occurring terms, topic models can provide insights into the underlying themes present in a collection of threat documents.

2.8.1 Latent Dirichlet Allocation

Latent Dirichlet Allocation (LDA) is one of the most widely used probabilistic topic modeling techniques.

LDA assumes that:

- Each document is associated with a mixture of topics.
- Each topic is represented by a probability distribution over words.
- Documents are generated through combinations of latent topics.

The probability of a document can be represented conceptually as a mixture of its underlying topics.

LDA has been applied to cybersecurity text to identify major themes and investigate changes in cyber threat discussions. Its interpretable topic-word distributions make it useful for exploratory analysis.

2.8.2 Non-Negative Matrix Factorization

Non-Negative Matrix Factorization (NMF) is another approach for extracting latent topics from document-term matrices.

Given a non-negative matrix X , NMF attempts to approximate it as:

$$X \approx WH$$

where W represents document-topic relationships and H represents topic-term relationships.

NMF can produce interpretable topics and can be applied to TF-IDF representations.

2.8.3 Dynamic and Temporal Topic Modeling

Traditional topic modeling generally treats a document collection as static. However, cyber threats evolve continuously. Temporal topic modeling addresses this limitation by considering the time associated with documents.

Documents can be grouped into time intervals such as:

- Daily
- Weekly
- Monthly
- Quarterly
- Yearly

Topics can then be extracted for different time periods and compared to identify changes in topic prevalence.

For example, if a topic related to a particular vulnerability, malware family, or attack technique shows a significant increase over successive periods, it may represent an emerging trend requiring further investigation.

Temporal topic analysis therefore provides an important foundation for the emerging-threat detection component of this research.

2.9 Emerging Threat Detection

Emerging threat detection refers to the identification of new, changing, or increasingly important cyber threat patterns. Unlike conventional classification, emerging-threat detection is concerned not only with known categories but also with changes that may indicate the development of new threats.

Cyber threats can emerge through several mechanisms, including:

- New vulnerabilities
- New malware families
- Changes in malware behavior
- New attack techniques
- New threat-actor campaigns
- Increasing exploitation of existing vulnerabilities
- Changes in attacker terminology
- New combinations of existing techniques

Traditional supervised classification can identify threats that belong to known classes, but it may not explicitly identify new patterns outside the training distribution.

Topic modeling provides a complementary approach. By monitoring topic distributions over time, it is possible to identify topics whose frequency, vocabulary, or relationships change substantially.

2.9.1 Temporal Trend Analysis

Let $T_k(t)$ represent the prevalence of topic k during time period t . A temporal trend can be represented as:

$$\text{Trend}_k(t) = T_k(t) - T_k(t-1)$$

A sustained increase in a topic's prevalence may indicate growing attention or activity associated with that topic.

However, an increase in a topic does not automatically imply the existence of a confirmed new cyber threat. Topic changes can also result from news coverage, repeated reporting, major security announcements, or changes in data collection. Therefore, emerging-topic detection should be treated as a signal-generation mechanism rather than definitive threat confirmation.

2.9.2 Emerging Threat Detection Using Topic Modeling

The base research associated with this thesis investigated the use of LDA and NMF for analyzing topic distributions over time and identifying emerging threats. Its findings demonstrate that topic modeling

can complement supervised threat classification by providing information about changing threat-related themes.

This research direction is particularly relevant because emerging threats may not have sufficient labeled examples for conventional supervised learning. Temporal topic modeling can instead identify changes in the distribution of textual themes without requiring every emerging pattern to have a predefined label.

2.10 Research Gap

The literature indicates that Machine Learning, Deep Learning, NLP, Transformer models, and topic modeling have each been applied to different aspects of cybersecurity and CTI analysis. However, several challenges remain.

2.10.1 Gap in Automated Classification and Emerging-Threat Analysis

Many supervised approaches focus primarily on assigning documents to predefined threat categories. Such systems can perform well for known classes but may provide limited information about previously unseen or emerging patterns.

2.10.2 Gap in Temporal Analysis

A considerable portion of CTI classification research treats documents independently or considers the dataset as a static collection. Cyber threats, however, change over time. A framework that explicitly considers temporal changes can provide additional information about threat evolution.

2.10.3 Gap in Integration of Classification and Topic Modeling

Threat classification and topic modeling are often considered separate analytical tasks. Classification provides predefined categories, while topic modeling discovers latent themes. Integrating both approaches can provide complementary information about known and evolving threats.

2.10.4 Gap in Handling Unstructured CTI

A significant amount of CTI is available in unstructured textual form. Extracting useful representations from such data remains challenging because cybersecurity text contains specialized terminology, identifiers, abbreviations, and domain-specific language.

2.10.5 Gap in Emerging Threat Identification

Existing classification models generally depend on historical labels. Consequently, they may have difficulty identifying new threat patterns that are not represented in the training dataset. Temporal topic analysis provides a possible approach for identifying changes in threat-related discussions and activity.

2.10.6 Gap in Timely CTI Analysis

The rapidly changing cybersecurity environment requires efficient processing of newly available information. Systems that require extensive manual preprocessing and analysis may introduce delays between the appearance of threat information and its interpretation.

2.10.7 Proposed Research Direction

Based on the identified gaps, this research proposes an integrated framework combining **Machine Learning-based cyber threat classification with temporal topic modeling for emerging threat detection**.

The proposed research aims to:

1. Process and normalize textual CTI data.
2. Extract meaningful textual features.
3. Classify CTI documents into predefined cyber threat categories.
4. Discover latent threat-related topics.
5. Track topic distributions across time.
6. Identify significant changes and potentially emerging threat patterns.
7. Evaluate classification and temporal-analysis performance.
8. Provide an integrated analytical framework for automated CTI analysis.

The proposed approach therefore addresses the gap between **known-threat classification** and **emerging-threat pattern detection** by combining supervised and unsupervised analytical techniques within a temporal CTI framework.

Chapter Summary

This chapter reviewed the major research areas relevant to automated Cyber Threat Intelligence analysis. The discussion covered CTI concepts and data sources, cyber threat classification, Natural Language Processing, Machine Learning, Deep Learning, Transformer models, topic modeling, and emerging-threat detection.

The literature demonstrates that Machine Learning can effectively support automated classification of known cyber threat categories, while topic modeling can help discover latent structures within unstructured CTI data. Temporal analysis provides an additional dimension for examining how these topics change over time.

The review also identified limitations in existing approaches, particularly the dependence on predefined categories, limited temporal analysis, and the separation between threat classification and emerging-threat detection. These observations provide the foundation for the methodology proposed in the subsequent chapters of this thesis.

Chapter 3 – Research Methodology

3.1 Proposed Methodology

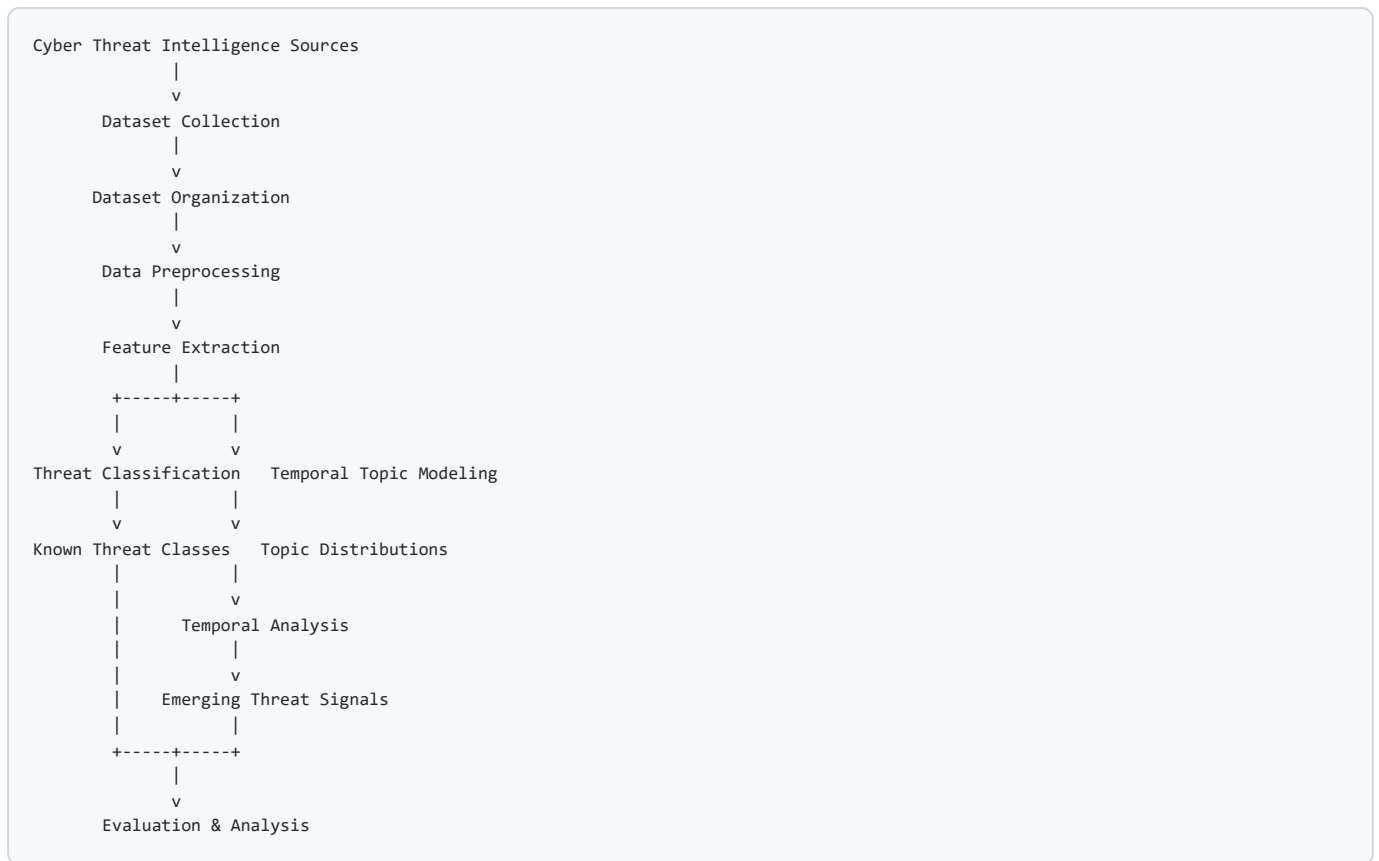
The proposed research methodology is designed to automate the analysis of Cyber Threat Intelligence (CTI) by combining Natural Language Processing (NLP), Machine Learning (ML), and temporal topic modeling techniques. The methodology focuses on two complementary objectives: **classification of known cyber threats** and **identification of potentially emerging threat patterns**.

The overall research workflow consists of the following major stages:

1. Dataset collection
2. Dataset integration and organization
3. Data preprocessing
4. Feature extraction and text representation
5. Cyber threat classification
6. Temporal topic modeling
7. Emerging threat detection
8. Performance evaluation
9. Comparative analysis and interpretation

The proposed methodology treats threat classification and emerging-threat detection as related but distinct analytical tasks. The classification component uses labeled CTI data to identify predefined threat categories, while the temporal topic modeling component analyzes the evolution of threat-related topics over time.

The overall workflow can be represented as:



The methodology is designed to support systematic analysis of textual CTI while maintaining a clear separation between supervised classification and unsupervised temporal analysis.

3.2 Dataset Collection

Dataset collection is the first major stage of the proposed methodology. The quality and relevance of the dataset directly influence the performance of the classification and topic modeling components.

The research focuses primarily on **textual Cyber Threat Intelligence reports and threat-related documents**. Such documents may contain information about malware, phishing, ransomware, vulnerabilities, denial-of-service attacks, exploit techniques, threat actors, campaigns, and other cybersecurity events.

3.2.1 Dataset Requirements

The dataset should contain sufficient textual information to support both supervised classification and temporal topic analysis.

The desired dataset fields include:

Field	Description
id	Unique identifier for each document
title	Title of the CTI report or document
text	Main textual content
category	Threat category or class label
date	Publication or observation date
source	Source of the CTI document
url	Reference URL where available

Not every source is required to provide all fields. When certain attributes are unavailable, the dataset may be standardized using the information available from the source.

3.2.2 Dataset Sources

Potential CTI data sources include:

- Cybersecurity reports
- Security advisories
- Malware analysis reports
- Vulnerability descriptions
- Cybersecurity incident reports
- Security blogs
- Threat intelligence publications
- Publicly available threat datasets
- Structured threat intelligence feeds

For this research, preference is given to datasets that provide both **textual information and temporal information**, because the proposed methodology requires documents to be analyzed according to their time of publication or observation.

3.2.3 Dataset Organization

Collected documents are combined into a standardized tabular format. Each document is treated as an individual observation.

A simplified representation is:

ID	Title	Text	Category	Date
001	Report A	Threat report...	Malware	2024-01-05
002	Report B	Security alert...	Phishing	2024-01-08
003	Report C	Attack report...	Ransomware	2024-01-14

The final dataset is divided into components required for supervised classification and temporal topic analysis.

3.2.4 Dataset Quality Considerations

Before model development, the dataset is examined for:

- Missing values
- Duplicate documents
- Invalid dates
- Empty text fields
- Incorrect or inconsistent labels
- Extremely short documents
- Formatting errors
- Class imbalance
- Temporal coverage

Data quality is particularly important because duplicate or incorrectly labeled documents can introduce bias into classification results and distort topic distributions.

3.3 Data Preprocessing

Raw CTI documents frequently contain irrelevant formatting, duplicated information, HTML elements, URLs, special characters, and cybersecurity-specific identifiers. Therefore, preprocessing is required before feature extraction and model training.

The preprocessing pipeline consists of several stages.

3.3.1 Data Cleaning

The collected dataset is first examined for incomplete and invalid records. Records with missing essential information may be removed or handled according to the availability of alternative information.

Duplicate documents are identified and removed to prevent repeated observations from influencing the model.

3.3.2 Text Normalization

Text normalization converts documents into a consistent representation.

Typical operations include:

- Converting text to lowercase where appropriate
- Removing unnecessary HTML tags
- Removing redundant whitespace
- Normalizing punctuation
- Removing irrelevant formatting
- Handling special characters
- Standardizing textual representations

However, cybersecurity-specific information should not be removed indiscriminately.

For example, identifiers such as:

```
CVE-2024-XXXX  
192.168.1.10  
malware-family-name  
example-domain.com
```

may contain useful cybersecurity information.

Therefore, preprocessing rules should preserve meaningful security-related tokens where they are relevant to the research task.

3.3.3 Tokenization

Tokenization divides a document into individual tokens.

For example:

```
Original:  
"Ransomware attacks targeted enterprise systems."  
  
Tokens:  
["ransomware", "attacks", "targeted", "enterprise", "systems"]
```

Tokenization provides the basic representation required by many NLP techniques.

3.3.4 Stop-Word Removal

Stop words are common words that generally provide limited discriminative information in traditional text classification.

Examples include:

```
the  
is  
and  
of  
to  
in
```

Removing stop words can reduce the dimensionality of the feature space.

However, domain-specific analysis should be used when deciding which words to remove because some common words may become meaningful in specific cybersecurity contexts.

3.3.5 Stemming and Lemmatization

Stemming reduces words to simplified root forms, whereas lemmatization attempts to convert words into their linguistically meaningful base forms.

For example:

```
attacking  
attacked  
attacks  
  
→ attack
```

Lemmatization may be preferred when maintaining semantic consistency is important.

3.3.6 Date Processing

Temporal information is essential for the proposed emerging-threat detection component.

Dates are therefore converted into a standardized format such as:

```
YYYY-MM-DD
```

Documents can subsequently be grouped into temporal intervals such as:

- Daily
- Weekly
- Monthly
- Quarterly

The selected interval depends on the volume and temporal distribution of the dataset.

3.3.7 Preprocessing Pipeline

The complete preprocessing workflow can be represented as:



3.4 Feature Extraction

Feature extraction converts preprocessed textual information into numerical representations that can be processed by Machine Learning algorithms.

Since ML models generally require numerical input, appropriate text representation is an essential component of the proposed methodology.

3.4.1 TF-IDF Representation

Term Frequency-Inverse Document Frequency (TF-IDF) is used as a primary traditional representation for textual CTI.

The TF-IDF value of a term t in document d can be expressed as:

$$TFIDF(t,d) = TF(t,d) \times IDF(t)$$

where:

$$IDF(t) = \log\left(\frac{N}{df(t)}\right)$$

Here:

- N represents the total number of documents.
- $df(t)$ represents the number of documents containing term t .
- $TF(t,d)$ represents the frequency of term t in document d .

TF-IDF assigns higher weights to terms that are important within a document but relatively uncommon across the complete corpus.

3.4.2 N-Gram Features

N-grams can be used to capture combinations of consecutive words.

For example:

```
Text:
"distributed denial service attack"

Unigrams:
distributed
denial
service
attack

Bigrams:
distributed denial
denial service
service attack
```

N-gram features can capture meaningful cybersecurity phrases that may not be represented adequately by individual words.

Examples include:

```
zero day
denial service
remote code execution
credential theft
```

3.4.3 Word Embeddings

Dense word representations such as Word2Vec and Doc2Vec can be used to represent semantic relationships between words or documents.

These approaches can provide richer representations than simple frequency-based methods. However, their effectiveness depends on the training corpus and the ability of the embeddings to capture cybersecurity-specific terminology.

3.4.4 Transformer-Based Representations

Transformer-based language models can generate contextual representations of CTI documents. These representations consider the surrounding context of words rather than treating each term independently.

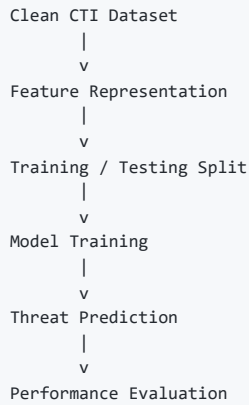
Transformer representations may be used as an advanced feature representation or as a comparative modeling approach.

The selection of feature representation depends on the experimental design, computational resources, and characteristics of the dataset.

3.5 Threat Classification

The threat classification component is designed to categorize CTI documents into predefined threat classes.

The classification process consists of:



3.5.1 Classification Categories

The exact categories depend on the available labeled dataset.

Possible categories include:

- Malware
- Phishing
- Ransomware
- DDoS
- Vulnerability Exploitation
- Credential Attacks
- Botnets
- Other cyber threats

Only categories supported by the selected dataset should be used in the final experimental implementation.

3.5.2 Machine Learning Algorithms

Several supervised ML algorithms may be evaluated.

Naïve Bayes

Naïve Bayes is a probabilistic classification algorithm that assumes conditional independence between features. It is computationally efficient and has traditionally been used for text classification.

Logistic Regression

Logistic Regression estimates the probability of a document belonging to a particular class. It is commonly used as a baseline for high-dimensional text classification.

Support Vector Machine

Support Vector Machine (SVM) attempts to identify a decision boundary that separates classes in feature space. SVM is particularly suitable for high-dimensional sparse representations such as TF-IDF.

Random Forest

Random Forest combines multiple decision trees to produce a classification decision. It can model nonlinear relationships and provides an ensemble-based alternative to linear classifiers.

The final selection of models should be based on experimental comparison rather than assuming that one algorithm will always perform best.

3.5.3 Training and Testing

The labeled dataset is divided into training and testing subsets.

The training data are used to learn the classification model, while the test data are reserved for evaluating its performance on previously unseen documents.

A stratified split may be used when class distributions are uneven so that the relative representation of classes is maintained across the subsets.

3.5.4 Classification Output

For an input CTI document d , the classification model produces a predicted class:

$$\hat{y} = f(d)$$

where:

- d is the input CTI document.
- f represents the trained classification model.
- $\hat{y}$ represents the predicted threat category.

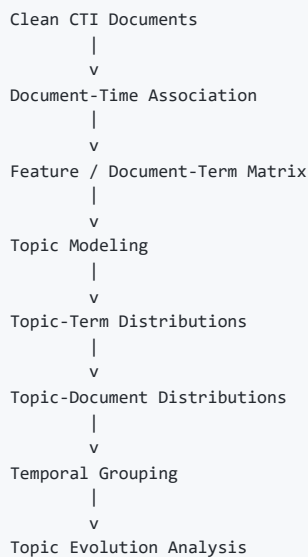
The predicted class can subsequently be used as an additional analytical attribute during CTI analysis.

3.6 Temporal Topic Modeling

Temporal topic modeling is used to discover latent themes in CTI data and investigate how those themes change over time.

Unlike supervised classification, topic modeling does not require every document to have a predefined threat label. This makes it useful for exploring patterns that may not correspond directly to known threat categories.

The temporal topic modeling pipeline consists of:



3.6.1 Document-Time Association

Each CTI document is associated with its publication or observation date.

For example:

```
Document 1 → January 2024
Document 2 → January 2024
Document 3 → February 2024
Document 4 → March 2024
```

This allows the corpus to be divided into temporal periods.

3.6.2 Latent Dirichlet Allocation

Latent Dirichlet Allocation (LDA) can be used to discover latent topics.

For a corpus containing K topics, LDA estimates:

- Topic distribution for each document.
- Word distribution for each topic.

The resulting topics can be interpreted by examining their highest-probability terms.

For example:

Topic 1:
ransomware, encryption, victim, extortion, payload

Topic 2:
phishing, email, credential, campaign, login

Topic 3:
vulnerability, exploit, patch, remote, execution

The actual topics will depend on the selected dataset and model parameters.

3.6.3 Non-Negative Matrix Factorization

NMF may also be used as an alternative topic modeling approach.

Given a document-term matrix X , NMF decomposes it into two non-negative matrices:

$$X \approx WH$$

where:

- W represents document-topic relationships.
- H represents topic-term relationships.

Comparing LDA and NMF can provide additional insight into the stability and interpretability of discovered topics.

3.6.4 Temporal Topic Distribution

After topics are extracted, their prevalence is calculated for different time periods.

For topic k and time period t , the topic prevalence can be represented as:

$$P(k,t) = \frac{\text{documents associated with topic } k \text{ during } t}{\text{total documents during } t}$$

The resulting values can be visualized as temporal trends.

For example:

Time →	T1	T2	T3	T4
Topic A	0.12	0.15	0.21	0.34
Topic B	0.30	0.27	0.22	0.18
Topic C	0.08	0.09	0.13	0.25

Such trends can be investigated for potential changes in the cyber threat landscape.

3.7 Emerging Threat Detection

The emerging-threat detection component uses temporal changes in CTI topics to identify potentially significant new or increasing patterns.

The objective is not to automatically declare every newly appearing topic as a confirmed cyber threat. Instead, the methodology generates **emerging-threat signals** that can be examined by analysts.

3.7.1 Topic Growth

For a topic k , its change between two consecutive periods can be calculated as:

$$\Delta P_k(t) = P(k,t) - P(k,t-1)$$

A positive value indicates that the relative prevalence of the topic has increased.

3.7.2 Growth Rate

A relative growth measure can also be calculated:

$$\text{GrowthRate}_k(t) = \frac{P(k,t) - P(k,t-1)}{P(k,t-1) + \epsilon}$$

where ϵ is a small value used to avoid division by zero.

This measure can help identify topics showing unusually large increases.

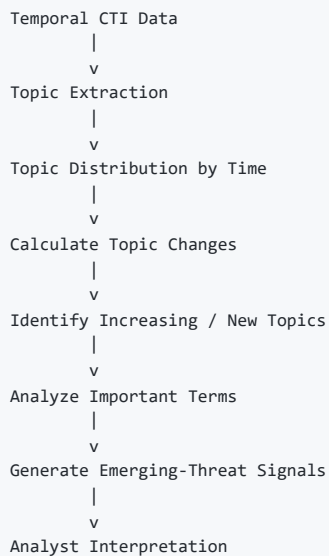
3.7.3 Emerging Topic Criteria

A topic may be considered a candidate emerging pattern when one or more of the following conditions are observed:

1. The topic appears for the first time in a later temporal period.
2. The topic shows a sustained increase in prevalence.
3. The topic demonstrates a substantial increase compared with previous periods.
4. New cybersecurity-specific terms become strongly associated with the topic.
5. The topic becomes increasingly associated with known threat categories or indicators.

These criteria are analytical signals rather than definitive evidence of a newly discovered attack.

3.7.4 Emerging Threat Detection Workflow



3.7.5 Relationship with Threat Classification

Threat classification and emerging-threat detection provide complementary information.

Classification answers:

“What known threat category does this document belong to?”

Temporal topic modeling addresses:

“What themes are changing or becoming increasingly prominent over time?”

The combination can therefore provide a broader analytical view of CTI.

For example, a group of documents may be classified as belonging to a known threat category while temporal topic analysis reveals that a new vulnerability, malware family, or attack technique is increasingly associated with those documents.

3.8 Evaluation Metrics

The proposed methodology requires separate evaluation procedures for threat classification and temporal topic modeling.

3.8.1 Accuracy

Accuracy measures the proportion of correctly classified observations among all observations.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

where:

- \$TP\$ = True Positive
- \$TN\$ = True Negative
- \$FP\$ = False Positive
- \$FN\$ = False Negative

Accuracy provides an overall measure of classification correctness but may be insufficient for imbalanced datasets.

3.8.2 Precision

Precision measures the proportion of predicted positive instances that are actually positive.

$$Precision = \frac{TP}{TP + FP}$$

Higher precision indicates fewer false-positive predictions.

3.8.3 Recall

Recall measures the proportion of actual positive instances that are correctly identified.

$$Recall = \frac{TP}{TP + FN}$$

Recall is particularly relevant when failing to identify a threat can have significant consequences.

3.8.4 F1-Score

F1-score is the harmonic mean of precision and recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

F1-score provides a balanced measure when both precision and recall are important.

3.8.5 Confusion Matrix

A confusion matrix provides a class-wise representation of classification results.

For a multiclass threat classification problem, the matrix can be represented as:

	Predicted			
	A	B	C	D
Actual A	TP	-	-	-
Actual B	-	TP	-	-
Actual C	-	-	TP	-
Actual D	-	-	-	TP

The confusion matrix can help identify which threat categories are frequently confused with one another.

3.8.6 Macro and Weighted Metrics

For multiclass classification, macro-averaged and weighted-averaged precision, recall, and F1-score may be reported.

Macro averaging gives equal importance to each class.

Weighted averaging considers the number of samples belonging to each class.

These metrics are useful when the dataset contains imbalanced threat categories.

3.8.7 Topic Coherence

Topic coherence can be used to evaluate the interpretability of discovered topics.

A higher coherence value generally indicates that the top words within a topic have stronger semantic or statistical relationships.

Topic coherence can be used to compare different numbers of topics or different topic modeling approaches.

3.8.8 Topic Diversity

Topic diversity measures how distinct the words used by different topics are.

High topic diversity indicates that different topics contain relatively different vocabularies rather than repeatedly using the same terms.

3.8.9 Temporal Trend Analysis

Emerging-threat detection can be evaluated using temporal characteristics such as:

- Topic frequency change
- Topic growth rate
- Topic persistence
- First appearance of topics
- Frequency of emerging-topic signals
- Relationship between emerging topics and known threat categories

Because an emerging topic is not necessarily a confirmed threat, these measures should primarily be interpreted as indicators of temporal change.

3.8.10 Comparative Evaluation

The final evaluation compares different classification models and, where applicable, different topic modeling configurations.

The comparison considers:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix
- Training and inference characteristics
- Topic coherence
- Topic diversity
- Temporal stability
- Interpretability

The results obtained from these evaluations are presented and analyzed in Chapter 5.

Chapter Summary

This chapter presented the proposed methodology for automated Cyber Threat Intelligence analysis. The methodology begins with CTI dataset collection and preparation, followed by text preprocessing and feature extraction.

Machine Learning algorithms are then used to classify CTI documents into predefined threat categories. In parallel, topic modeling is applied to discover latent themes, and temporal analysis is used to investigate how these themes evolve over time.

The emerging-threat detection component analyzes changes in topic prevalence and related textual patterns to generate candidate emerging-threat signals. Finally, classification and topic-modeling performance are evaluated using appropriate quantitative and analytical measures.

The methodology establishes the foundation for the experimental implementation and results presented in the subsequent chapters.

Chapter 4 – Proposed Framework and Implementation

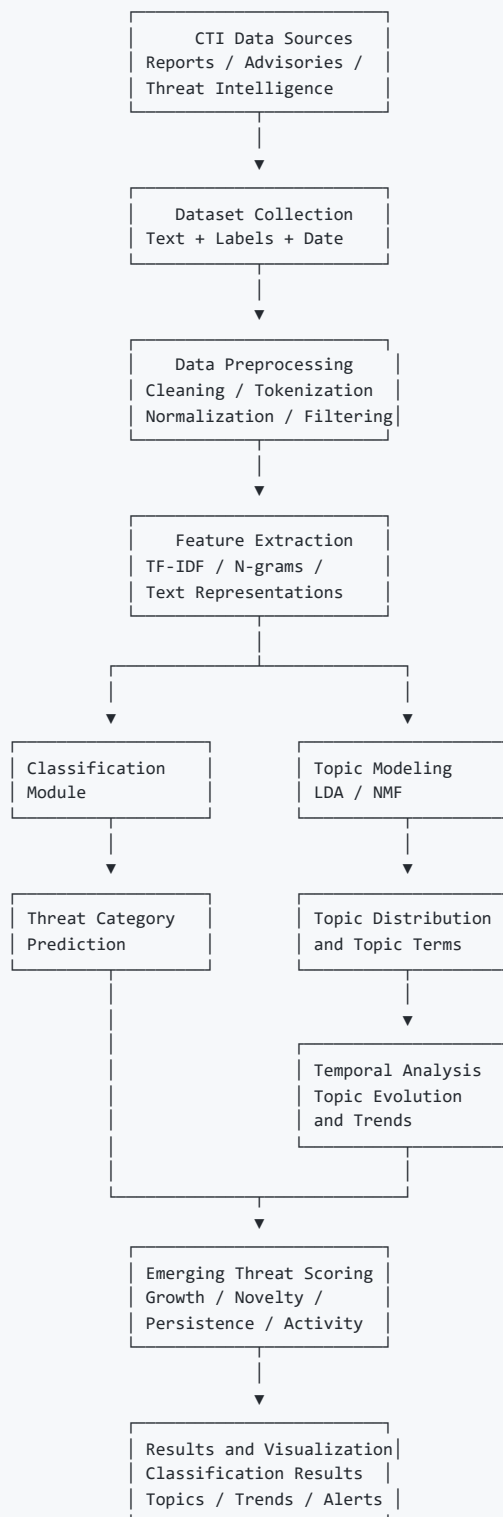
4.1 System Architecture

The proposed framework is designed to automate the analysis of Cyber Threat Intelligence (CTI) reports by combining machine learning-based threat classification with temporal topic modeling and emerging threat detection. The framework processes textual CTI data, extracts meaningful features, classifies threat-related documents, identifies latent topics, analyzes their temporal evolution, and assigns an emerging threat score to potentially significant patterns.

The architecture is organized into several interconnected stages:

1. CTI Data Input
2. Data Preprocessing
3. Feature Extraction
4. Threat Classification
5. Topic Modeling
6. Temporal Analysis
7. Emerging Threat Scoring
8. Results and Visualization

The overall architecture of the proposed framework is represented below.



4.1.1 Architecture Components

The proposed architecture consists of the following major components.

CTI Data Sources

The input layer contains textual cyber threat information obtained from publicly available CTI reports, security advisories, incident reports, vulnerability information, malware reports, and other suitable threat intelligence sources.

Each document should contain, where available, textual information and temporal metadata such as publication date.

Dataset Management Layer

The collected documents are organized into a structured dataset. Important fields may include:

Field	Description
id	Unique document identifier
title	Title of the CTI document
text	Main textual content
category	Threat category or class
date	Publication or observation date
source	Source of the document
url	Reference URL

Preprocessing Layer

The preprocessing component removes irrelevant content and converts raw CTI documents into a consistent representation suitable for machine learning and topic modeling.

Feature Extraction Layer

The cleaned text is transformed into numerical representations. TF-IDF and n-gram features can be used for traditional machine learning models, while other text representations may be evaluated where required.

Classification Layer

The classification module predicts the threat category associated with each CTI document. Multiple machine learning algorithms can be trained and compared.

Topic Modeling Layer

The topic modeling component discovers latent themes from CTI documents without requiring predefined topic labels. LDA and NMF can be used to identify important cybersecurity topics.

Temporal Analysis Layer

Topic distributions are grouped according to time periods to identify changes in threat-related discussions.

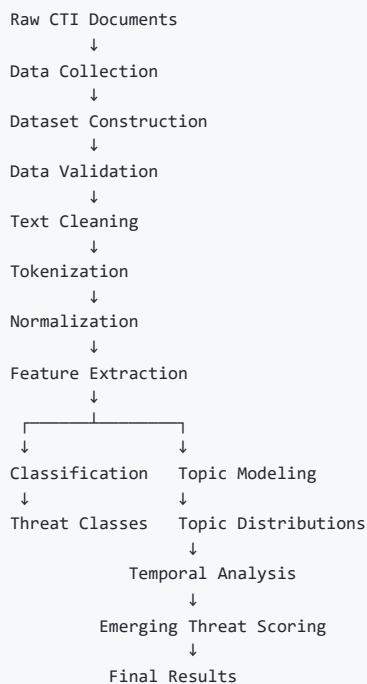
Emerging Threat Scoring Layer

The framework combines temporal and textual signals to identify topics that exhibit characteristics associated with emerging threats.

4.2 Data Pipeline

The data pipeline defines the sequence through which CTI documents are transformed from raw textual information into classification results and emerging threat indicators.

The pipeline consists of the following stages:



4.2.1 Data Collection

The first stage involves collecting CTI documents from selected sources. The collected documents are stored with their associated metadata.

The dataset should maintain the relationship between each document and its publication or observation date because temporal information is essential for emerging threat analysis.

4.2.2 Data Validation

Before preprocessing, the collected dataset is examined for:

- Missing textual content
- Missing dates
- Duplicate documents
- Invalid records
- Extremely short documents
- Incorrect labels
- Inconsistent formatting
- Duplicate URLs
- Unusable or corrupted records

Records that do not satisfy the required quality conditions are removed or corrected where appropriate.

4.2.3 Text Processing

The textual content is cleaned using the preprocessing operations described in Chapter 3. The objective is to reduce irrelevant variation while preserving cybersecurity-specific information.

Particular attention is given to security-related entities such as:

- CVE identifiers
- Malware names
- Vulnerability names
- Attack techniques
- Protocol names
- File extensions
- Domains
- IP addresses
- Security tools
- Threat actor names

Over-aggressive preprocessing can remove information that is useful for CTI analysis. Therefore, preprocessing rules are selected according to the characteristics of the dataset.

4.2.4 Data Representation

After preprocessing, documents are converted into numerical representations.

For traditional machine learning classification, TF-IDF is used to represent the importance of terms within documents.

For topic modeling, the corpus is represented using an appropriate document-term matrix.

4.2.5 Parallel Analytical Processing

After feature extraction, the framework follows two analytical paths.

The first path performs supervised threat classification.

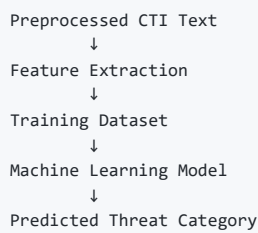
The second path performs unsupervised topic discovery and temporal analysis.

The outputs of both paths are subsequently used for integrated threat analysis.

4.3 Classification Module

The classification module is responsible for automatically assigning CTI documents to predefined threat categories.

The classification process can be represented as:



4.3.1 Classification Objective

Let the training dataset be represented as:

$$D = \{(x_i, y_i)\}_{i=1}^N$$

where:

- x_i represents the feature vector of document i ,
- y_i represents its corresponding threat category,
- N represents the total number of documents.

The classification model learns a function:

$$f: X \rightarrow Y$$

where X represents the document feature space and Y represents the set of predefined threat categories.

For a new CTI document x , the trained model generates:

$$\hat{y} = f(x)$$

where $\hat{y}$ is the predicted threat category.

4.3.2 Classification Models

The framework can evaluate multiple machine learning algorithms, including:

- Naive Bayes
- Logistic Regression
- Support Vector Machine (SVM)
- Random Forest
- Gradient Boosting

The final set of models depends on dataset characteristics and experimental requirements.

4.3.3 Training and Testing

The dataset is divided into training and testing subsets.

A typical configuration may use:

- Training set: 80%
- Testing set: 20%

A stratified split should be considered when the dataset contains class imbalance.

The training data are used to learn the model parameters, while the test data are kept separate for evaluating generalization performance.

4.3.4 Classification Output

For each CTI document, the classification module produces an output similar to:

Document ID	Predicted Category	Confidence/Probability
CTI-001	Malware	0.91
CTI-002	Phishing	0.87
CTI-003	Ransomware	0.94

The actual output format depends on the selected machine learning implementation.

Classification results are evaluated using accuracy, precision, recall, F1-score, and confusion matrices.

4.3.5 Classification and Emerging Threat Analysis

The classification module and emerging threat module perform different functions.

Classification determines **what predefined threat category a document belongs to**, whereas emerging threat analysis attempts to identify **new or increasingly prominent patterns over time**.

Therefore, a topic can be considered important for emerging-threat analysis even when it does not correspond directly to a newly created classification category.

4.4 Topic Modeling Module

The topic modeling module discovers latent thematic structures in the CTI corpus.

Unlike supervised classification, topic modeling does not require predefined topic labels. It attempts to identify groups of terms that frequently occur together across documents.

The proposed framework can use:

- Latent Dirichlet Allocation (LDA)
- Non-Negative Matrix Factorization (NMF)

4.4.1 LDA-Based Topic Modeling

LDA represents documents as mixtures of topics and topics as probability distributions over words.

For a document d , the topic distribution can be represented as:

$$\theta_d = P(z|d)$$

where z represents a latent topic.

Similarly, each topic can be represented by a word distribution:

$$\phi_k = P(w|z_k)$$

where:

- w represents a word,
- z_k represents topic k .

The highest-probability words associated with each topic are extracted to interpret the semantic meaning of the topic.

4.4.2 NMF-Based Topic Modeling

NMF decomposes a non-negative document-term matrix into two lower-dimensional matrices.

Let X represent the document-term matrix:

$$X \approx WH$$

where:

- W represents document-topic weights,
- H represents topic-term weights.

The resulting topic-term matrix can be used to identify the most representative words associated with each topic.

4.4.3 Topic Identification

For each discovered topic, the framework extracts the most representative terms.

An example output is:

Topic	Representative Terms	Possible Interpretation
Topic 1	ransomware, encryption, file, ransom	Ransomware activity
Topic 2	phishing, email, credential, link	Phishing activity
Topic 3	vulnerability, exploit, patch, CVE	Vulnerability exploitation
Topic 4	botnet, command, server, infected	Botnet activity

The interpretations shown in such a table are analytical interpretations of the discovered terms and should be validated against the underlying documents.

4.4.4 Topic Number Selection

The number of topics is an important hyperparameter.

Different values of K can be evaluated using measures such as:

- Topic coherence
- Topic diversity
- Interpretability
- Temporal stability
- Domain relevance

The selected number of topics should therefore be justified using both quantitative and qualitative analysis.

4.5 Temporal Analysis

Temporal analysis examines how CTI topics change over time.

The primary objective is to determine whether particular threat-related topics:

- Appear for the first time
- Increase in prevalence
- Decrease in prevalence
- Persist over multiple periods
- Reappear after periods of low activity
- Develop new associated terminology

4.5.1 Temporal Grouping

Each CTI document is associated with a date.

The dataset can be divided into temporal intervals such as:

- Daily
- Weekly
- Monthly
- Quarterly

The selected interval depends on dataset size and temporal coverage.

For an interval t , the prevalence of topic k can be represented as:

$$P(k,t)$$

$$\frac{\sum_{d \in D_t} \theta_{d,k}}{|D_t|}$$

where:

- D_t represents documents belonging to time interval t ,
- $\theta_{d,k}$ represents the contribution of topic k to document d ,
- $P(k,t)$ represents the average prevalence of topic k during interval t .

4.5.2 Topic Evolution

The change in topic prevalence between consecutive periods can be calculated as:

$$\Delta P_k(t)$$

$$P(k,t) - P(k,t-1)$$

A positive value indicates an increase in topic prevalence, while a negative value indicates a decrease.

A normalized growth rate can also be calculated as:

\$\$ G_k(t)

$$\frac{P(k,t)-P(k,t-1)}{P(k,t-1)+\epsilon}$$

where ϵ is a small positive constant used to avoid division by zero.

4.5.3 Temporal Topic Matrix

The temporal topic distribution can be organized into a matrix:

Time Period	Topic 1	Topic 2	Topic 3	Topic 4
Period 1	0.18	0.25	0.31	0.26
Period 2	0.21	0.23	0.29	0.27
Period 3	0.34	0.19	0.25	0.22
Period 4	0.42	0.17	0.22	0.19

Such a representation allows changes in threat-related topics to be examined systematically.

4.5.4 Temporal Visualization

The temporal analysis can be visualized using:

- Topic prevalence line charts
- Topic heatmaps
- Time-series plots
- Topic evolution graphs
- Word-frequency trends

These visualizations help identify periods of unusual increases in particular threat-related themes.

4.6 Emerging Threat Scoring

The emerging threat scoring module is used to prioritize topics that demonstrate characteristics associated with emerging threat activity.

An important distinction is maintained between an **emerging threat signal** and a **confirmed new or zero-day threat**. A high score indicates that a topic satisfies selected analytical criteria; it does not independently establish that the underlying activity represents a confirmed security incident or zero-day vulnerability.

4.6.1 Scoring Factors

The proposed scoring mechanism considers multiple signals:

1. **Novelty**
2. **Growth**
3. **Persistence**
4. **Activity**
5. **Term Change**

These factors are intended to capture different dimensions of emerging behavior.

4.6.2 Novelty Score

Novelty represents the extent to which a topic is newly appearing in the observed dataset.

A simple binary representation can be defined as:

$$N_k(t) = \begin{cases} 1, & \text{if topic } k \text{ is newly observed at } t \\ 0, & \text{otherwise} \end{cases}$$

More detailed novelty measures can be developed using the appearance of new topic terms or changes in topic distributions.

4.6.3 Growth Score

The growth component measures the increase in topic prevalence:

$$G_k(t)$$

$$\frac{P(k,t) - P(k,t-1)}{P(k,t-1) + \epsilon}$$

Large positive values indicate rapid increases relative to the previous period.

Because raw growth values may have different ranges, they can be normalized before combining them with other components.

4.6.4 Persistence Score

Persistence measures whether a topic remains active over multiple consecutive time periods.

For example:

$$S_k(t) = \frac{C_k(t)}{W}$$

where:

- $C_k(t)$ is the number of recent time windows in which topic k remained active,
- W is the total number of considered windows.

A persistent topic receives a higher score than a topic that appears only once.

4.6.5 Activity Score

The activity component represents the relative amount of CTI information associated with a topic.

It can be estimated using the number or proportion of documents assigned strongly to the topic during a given time period.

4.6.6 Term Change Score

The framework can also examine changes in important terms associated with a topic.

The appearance of previously uncommon cybersecurity terms, malware names, attack techniques, vulnerabilities, or infrastructure-related terminology can provide additional evidence for a changing topic.

4.6.7 Combined Emerging Threat Score

The individual components can be combined into a normalized score:

$ETS_k(t)$

$$w_{NN}k(t) + w_{GG}k(t) + w_{SS}k(t) + w_{AA}k(t) + w_{TT}k(t)$$

where:

- $ETS_k(t)$ is the Emerging Threat Score,
- $N_k(t)$ is the novelty score,
- $G_k(t)$ is the growth score,
- $S_k(t)$ is the persistence score,
- $A_k(t)$ is the activity score,
- $T_k(t)$ is the term-change score,
- w_N, w_G, w_S, w_A, w_T are weighting coefficients.

The weights satisfy:

$$w_N + w_G + w_S + w_A + w_T = 1$$

The weights can initially be selected based on the experimental design and subsequently examined through sensitivity analysis.

4.6.8 Emerging Threat Signal

A topic may be flagged for further investigation when its score exceeds a selected threshold:

$$ETS_k(t) \geq \tau$$

where τ represents the selected threshold.

The threshold should be determined experimentally rather than arbitrarily. The resulting output should be interpreted as a **candidate emerging-threat signal** requiring further security validation.

An example output is:

Topic	Time Period	Growth	Persistence	Novelty	ETS	Status
Topic 3	Period 2	0.21	0.70	0.80	0.61	Candidate
Topic 7	Period 3	0.48	0.85	0.75	0.79	Candidate
Topic 2	Period 4	0.05	0.30	0.10	0.12	Low Signal

The values in this table are illustrative and will be replaced by experimentally obtained values in Chapter 5.

4.7 Implementation

The proposed framework is implemented using a Python-based data science environment. The implementation is divided into data handling, preprocessing, machine learning, topic modeling, temporal analysis, scoring, and visualization components.

4.7.1 Software Environment

The implementation can use the following software components:

Component	Technology
Programming Language	Python
Development Environment	Jupyter Notebook
Data Processing	Pandas, NumPy
Text Processing	NLTK / spaCy
Feature Extraction	scikit-learn
Machine Learning	scikit-learn
Topic Modeling	scikit-learn / Gensim
Visualization	Matplotlib / Seaborn
Data Storage	CSV / JSON / SQLite
Version Control	Git

The exact libraries and versions used for the final experiments should be documented to support reproducibility.

4.7.2 Implementation Workflow

The implementation follows the sequence:

1. Load CTI Dataset
↓
2. Validate Dataset
↓
3. Clean and Preprocess Text
↓
4. Generate TF-IDF Features
↓
5. Split Dataset
↓
6. Train Classification Models
↓
7. Evaluate Classification
↓
8. Build Topic Modeling Corpus
↓
9. Train LDA / NMF Models
↓
10. Extract Topic Distributions
↓
11. Group Topics by Time
↓
12. Calculate Temporal Changes
↓
13. Calculate Emerging Threat Scores
↓
14. Generate Visualizations
↓
15. Store and Analyze Results

4.7.3 Dataset Loading

The dataset is loaded into a structured data frame.

A simplified implementation is:

```
import pandas as pd

df = pd.read_csv("cti_reports.csv")

print(df.shape)
print(df.columns)
print(df.head())
```

The dataset is then checked for missing values and duplicate records.

```
print(df.isnull().sum())
print("Duplicate records:", df.duplicated().sum())
```

4.7.4 Text Preprocessing

A preprocessing function is applied to the textual data.

```
import re

def clean_text(text):
    text = str(text).lower()
    text = re.sub(r"\s+", " ", text)
    text = re.sub(r"^[a-zA-Z0-9\s\-\._:/]", " ", text)
    return text.strip()

df["clean_text"] = df["text"].apply(clean_text)
```

The final preprocessing implementation may be extended with tokenization, stop-word removal, lemmatization, and cybersecurity-specific normalization according to the selected dataset.

4.7.5 TF-IDF Feature Extraction

TF-IDF features are generated using scikit-learn.

```
from sklearn.feature_extraction.text import TfidfVectorizer

vectorizer = TfidfVectorizer(
    max_features=10000,
    ngram_range=(1, 2),
    min_df=2
)

X = vectorizer.fit_transform(df["clean_text"])
```

The final hyperparameters should be selected through experimentation and documented in Chapter 5.

4.7.6 Classification Implementation

The classification models can be implemented using scikit-learn.

```
from sklearn.model_selection import train_test_split
from sklearn.svm import LinearSVC

X_train, X_test, y_train, y_test = train_test_split(
    X,
    df["category"],
    test_size=0.20,
    random_state=42,
    stratify=df["category"]
)

model = LinearSVC()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
```

Other algorithms can be implemented using the same training and evaluation pipeline.

4.7.7 Topic Modeling Implementation

For NMF-based topic modeling:

```

from sklearn.decomposition import NMF

nmf = NMF(
    n_components=10,
    random_state=42
)

W = nmf.fit_transform(X)
H = nmf.components_

```

The most representative terms for each topic can then be extracted.

```

terms = vectorizer.get_feature_names_out()

for topic_idx, topic in enumerate(H):
    top_terms = topic.argsort()[-10:][::-1]
    words = [terms[i] for i in top_terms]
    print(f"Topic {topic_idx + 1}: {words}")

```

LDA can similarly be implemented using an appropriate document-term representation.

4.7.8 Temporal Topic Analysis

After obtaining document-topic representations, the topic scores are associated with the document dates.

```

df["date"] = pd.to_datetime(df["date"])

topic_columns = [
    f"topic_{i+1}"
    for i in range(W.shape[1])
]

topic_df = pd.DataFrame(
    W,
    columns=topic_columns,
    index=df.index
)

df = pd.concat([df, topic_df], axis=1)

```

The data can then be grouped by month or another selected time interval.

```

df["period"] = df["date"].dt.to_period("M")

temporal_topics = df.groupby("period")[topic_columns].mean()

```

This produces a temporal topic matrix suitable for trend analysis.

4.7.9 Emerging Threat Score Implementation

The temporal topic matrix can be used to calculate changes between consecutive periods.

```

topic_growth = temporal_topics.pct_change()

```

Additional components such as persistence, novelty, and activity can then be normalized and combined according to the scoring formulation defined in Section 4.6.

A conceptual implementation is:

```
emerging_score = (  
    w_n * novelty_score +  
    w_g * growth_score +  
    w_s * persistence_score +  
    w_a * activity_score +  
    w_t * term_change_score  
)
```

The final implementation should ensure that all component scores are normalized to comparable ranges before calculating the combined score.

4.7.10 Visualization

Visualization is used to support interpretation of classification and temporal results.

Important visualizations include:

- Class distribution plots
- Confusion matrices
- Precision-recall curves
- F1-score comparison
- Topic-word distributions
- Topic prevalence over time
- Temporal heatmaps
- Emerging threat score trends

For example, topic evolution can be visualized using a line plot:

```
import matplotlib.pyplot as plt  
  
temporal_topics.plot(figsize=(12, 6))  
plt.xlabel("Time Period")  
plt.ylabel("Average Topic Contribution")  
plt.title("Temporal Evolution of CTI Topics")  
plt.legend()  
plt.tight_layout()  
plt.show()
```

4.7.11 Result Storage

The framework stores intermediate and final results to support reproducibility.

Potential output files include:

```
results/
├── classification/
│   ├── classification_report.csv
│   ├── confusion_matrix.csv
│   └── model_comparison.csv
├── topics/
│   ├── topic_terms.csv
│   ├── document_topics.csv
│   └── topic_coherence.csv
├── temporal/
│   ├── temporal_topic_distribution.csv
│   └── topic_growth.csv
├── emerging/
│   ├── emerging_threat_scores.csv
│   └── candidate_emerging_topics.csv
└── figures/
    ├── class_distribution.png
    ├── confusion_matrix.png
    ├── topic_distribution.png
    └── temporal_trends.png
```

4.7.12 Reproducibility

To ensure reproducibility, the implementation records:

- Dataset version
- Dataset preprocessing procedure
- Feature extraction parameters
- Model parameters
- Topic number
- Random seed
- Train-test configuration
- Scoring weights
- Emerging threat threshold
- Python and library versions

Random states should be fixed where applicable so that experiments can be reproduced.

Chapter Summary

This chapter presented the proposed framework and its implementation for automated CTI analysis. The framework integrates data preprocessing, feature extraction, machine learning-based threat classification, topic modeling, temporal analysis, and emerging threat scoring.

The classification module identifies predefined threat categories, while the topic modeling module discovers latent threat-related themes. Temporal analysis tracks changes in these themes over time, and the emerging threat scoring mechanism combines novelty, growth, persistence, activity, and term-change signals to identify candidate emerging threat patterns.

The implementation provides a practical computational framework for evaluating the proposed research objectives. The experimental configuration, dataset distribution, model performance, topic quality, temporal trends, and emerging threat results are presented and analyzed in Chapter 5.

Chapter 5 – Experimental Results and Analysis

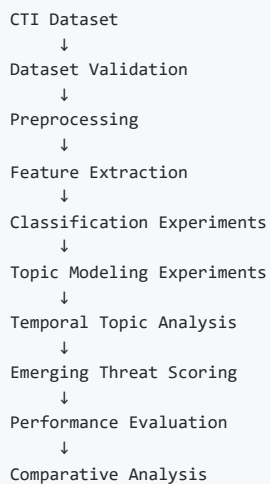
5.1 Experimental Setup

The experimental evaluation was conducted to assess the performance of the proposed Cyber Threat Intelligence (CTI) analysis framework. The framework integrates machine learning-based threat classification, topic modeling, temporal analysis, and emerging threat scoring.

The experiments were designed to evaluate the following aspects:

1. Quality and distribution of the collected CTI dataset.
2. Performance of machine learning models for threat classification.
3. Comparative performance of different classification algorithms.
4. Quality and interpretability of discovered topics.
5. Temporal evolution of cybersecurity-related topics.
6. Identification of candidate emerging threat patterns.
7. Effectiveness of the proposed emerging threat scoring mechanism.
8. Computational and practical limitations of the framework.

The experimental workflow follows the sequence:



The experiments were performed using the implementation described in Chapter 4. The exact dataset version, preprocessing configuration, model parameters, and software versions used for the final experiments are documented to ensure reproducibility.

5.1.1 Experimental Configuration

The major experimental parameters are summarized below.

Parameter	Configuration
Programming Language	Python
Text Representation	TF-IDF
N-gram Range	1–2
Train-Test Split	80:20
Random State	42
Classification Models	Naive Bayes, Logistic Regression, SVM, Random Forest
Topic Models	LDA, NMF
Temporal Unit	Based on dataset coverage
Evaluation Metrics	Accuracy, Precision, Recall, F1-score
Topic Evaluation	Coherence, Topic Diversity, Interpretability
Visualization	Matplotlib / Seaborn
Development Environment	Jupyter Notebook

The final values should be updated according to the actual configuration used in the experiments.

5.2 Hardware and Software Requirements

The proposed framework requires a general-purpose computing environment capable of processing textual datasets and training machine learning and topic modeling models.

5.2.1 Hardware Requirements

The minimum and recommended configurations are presented below.

Component	Requirement
Processor	Modern multi-core CPU
RAM	Minimum 8 GB; 16 GB or more recommended
Storage	Minimum 10 GB available space
GPU	Optional for traditional ML and LDA/NMF
Operating System	Windows / Linux / macOS

For the proposed experiments, CPU-based execution is sufficient for traditional machine learning, TF-IDF feature extraction, LDA, and NMF for moderate-sized datasets.

If transformer-based representations or large-scale deep learning models are incorporated in future experiments, GPU acceleration may be beneficial.

5.2.2 Software Requirements

Software	Purpose
Python	Primary programming language
Jupyter Notebook	Experimental development
Pandas	Dataset processing
NumPy	Numerical computation
scikit-learn	Machine learning and feature extraction
NLTK / spaCy	Natural Language Processing
Gensim	Topic modeling where applicable
Matplotlib	Visualization
Seaborn	Statistical visualization
Git	Version control

The exact software versions should be recorded in the final experimental environment.

5.3 Dataset Distribution Analysis

The dataset distribution was analyzed before model training to understand its size, class composition, temporal coverage, and potential imbalance.

5.3.1 Dataset Size

The final dataset contains:

- Total documents: [N]
- Training documents: [N_train]
- Testing documents: [N_test]
- Number of threat categories: [C]
- Number of temporal periods: [T]

Replace the bracketed values with the actual experimental results.

5.3.2 Class Distribution

The distribution of documents across threat categories is an important factor in supervised classification.

Threat Category	Number of Documents	Percentage
Malware	[N]	[X]%
Phishing	[N]	[X]%
Ransomware	[N]	[X]%
DDoS	[N]	[X]%
Vulnerability Exploitation	[N]	[X]%
Other	[N]	[X]%
Total	[N]	100%

The final categories should correspond to the labels available in the selected dataset.

5.3.3 Class Imbalance

Class imbalance occurs when some threat categories contain substantially more documents than others.

The class distribution was examined using:

- Number of documents per class.
- Percentage contribution of each class.
- Majority-to-minority class ratio.
- Macro and weighted evaluation metrics.

If substantial imbalance exists, accuracy alone may not provide a complete representation of model performance. Therefore, precision, recall, and F1-score are also considered.

5.3.4 Temporal Distribution

The temporal distribution of documents was analyzed to determine the availability of CTI information across different periods.

Time Period	Number of Documents
Period 1	[N]
Period 2	[N]
Period 3	[N]
Period 4	[N]
...	...

The temporal distribution is particularly important because the proposed emerging-threat analysis depends on observing changes across time.

5.3.5 Dataset Quality Analysis

The dataset was examined for:

- Missing values.
- Duplicate records.
- Empty documents.
- Invalid dates.
- Incorrect labels.
- Extremely short documents.
- Duplicate textual content.
- Inconsistent formatting.

The number of records removed during preprocessing should be reported.

Data Quality Check	Records Identified	Records Removed/Handled
Missing text	[N]	[N]
Missing date	[N]	[N]
Duplicate records	[N]	[N]
Empty documents	[N]	[N]
Invalid labels	[N]	[N]

5.4 Final Dataset Format

After data cleaning and preprocessing, the CTI dataset was converted into a structured format suitable for classification and temporal topic analysis.

The final dataset contains the following fields:

Field	Data Type	Description
id	String	Unique document identifier
title	String	CTI document title
text	Text	Original or extracted CTI content
clean_text	Text	Preprocessed text
category	Categorical	Threat class
date	Date	Publication/observation date
source	String	Source of CTI information
url	String	Reference URL
period	Categorical	Temporal grouping

The cleaned text is used for feature extraction, while the date and period fields are used for temporal topic analysis.

5.5 Dataset Availability

The dataset used in the experimental study should be documented according to its source, licensing conditions, and reproducibility requirements.

Where redistribution is permitted, the processed dataset may be included with the project repository. If redistribution is restricted, the thesis should provide:

- Dataset source.
- Dataset collection procedure.
- Collection period.
- Number of documents.
- Preprocessing procedure.
- Dataset schema.
- Relevant source references.

The dataset should not contain sensitive personal information or restricted information that cannot legally or ethically be redistributed.

The final thesis should clearly distinguish between:

1. Original publicly available data.
2. Data collected by the researcher.
3. Preprocessed data.
4. Experiment-generated features and results.

5.6 Model Implementation

The classification and topic modeling experiments were implemented using the workflow described in Chapter 4.

5.6.1 Feature Extraction

TF-IDF was used to transform preprocessed CTI text into numerical feature vectors.

The resulting matrix is represented as:

$$X \in \mathbb{R}^{N \times V}$$

where:

- N represents the number of documents.
- V represents the vocabulary size.

The feature matrix was subsequently supplied to the classification algorithms.

5.6.2 Classification Models

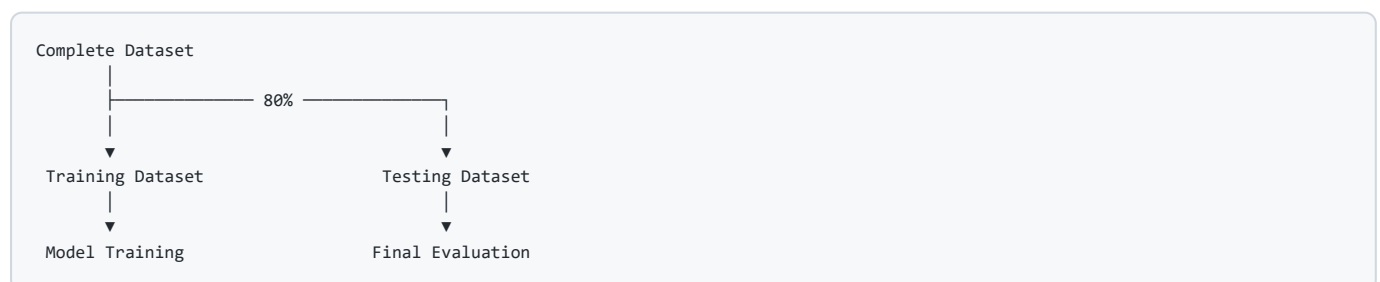
The following models were evaluated:

1. Naive Bayes
2. Logistic Regression
3. Support Vector Machine
4. Random Forest

The objective was not only to evaluate individual models but also to understand how different algorithms behave on CTI text classification.

5.6.3 Training Configuration

The dataset was divided into training and testing subsets.



Where appropriate, stratified sampling was used to maintain approximately similar class proportions between training and testing sets.

5.7 Classification Performance

The performance of the classification models was evaluated using accuracy, precision, recall, and F1-score.

5.7.1 Accuracy

Accuracy measures the proportion of correctly classified documents.

Accuracy = $\frac{TP+TN}{TP+TN+FP+FN}$

For multiclass classification, accuracy represents the proportion of correctly predicted samples across all classes.

5.7.2 Precision

Precision measures the proportion of predicted positive samples that are actually positive.

Precision = $\frac{TP}{TP+FP}$

5.7.3 Recall

Recall measures the proportion of actual positive samples correctly identified by the model.

Recall = $\frac{TP}{TP+FN}$

5.7.4 F1-Score

The F1-score is the harmonic mean of precision and recall.

F1 = $2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

For multiclass classification, macro-average and weighted-average F1-scores can be reported.

5.7.5 Overall Classification Results

The final classification results should be reported in the following format.

Model	Accuracy	Precision	Recall	F1-Score
Naive Bayes	[XX.XX]%	[XX.XX]%	[XX.XX]%	[XX.XX]%
Logistic Regression	[XX.XX]%	[XX.XX]%	[XX.XX]%	[XX.XX]%
SVM	[XX.XX]%	[XX.XX]%	[XX.XX]%	[XX.XX]%
Random Forest	[XX.XX]%	[XX.XX]%	[XX.XX]%	[XX.XX]%

The values should be replaced with the actual experimental measurements.

5.8 Comparative Model Analysis

The classification models were compared using their performance across the selected evaluation metrics.

The comparison focuses on:

- Overall accuracy.
- Precision.
- Recall.
- F1-score.
- Class-wise performance.
- Robustness to class imbalance.
- Computational requirements.

A comparative table can be used to summarize the results.

Model	Accuracy	Macro F1	Weighted F1	Training Time
Naive Bayes	[XX.XX]	[XX.XX]	[XX.XX]	[XX] sec
Logistic Regression	[XX.XX]	[XX.XX]	[XX.XX]	[XX] sec
SVM	[XX.XX]	[XX.XX]	[XX.XX]	[XX] sec
Random Forest	[XX.XX]	[XX.XX]	[XX.XX]	[XX] sec

The comparison should be interpreted according to the experimental results rather than assuming that a particular algorithm will perform best before experimentation.

5.8.1 Class-Wise Performance

Class-wise evaluation provides additional information about categories that are easier or more difficult to classify.

Threat Category	Precision	Recall	F1-Score	Support
Category 1	[XX.XX]	[XX.XX]	[XX.XX]	[N]
Category 2	[XX.XX]	[XX.XX]	[XX.XX]	[N]
Category 3	[XX.XX]	[XX.XX]	[XX.XX]	[N]
Category 4	[XX.XX]	[XX.XX]	[XX.XX]	[N]

Differences between classes may result from differences in sample size, terminology, topic overlap, and the similarity of threat descriptions.

5.9 Confusion Matrix Analysis

A confusion matrix was generated to analyze the classification behavior of the selected models.

For a multiclass classification problem, the confusion matrix is represented as:

$$C_{ij}$$

Number of samples belonging to class i predicted as class j

An example structure is:

Actual / Predicted	Class A	Class B	Class C	Class D
Class A	[N]	[N]	[N]	[N]
Class B	[N]	[N]	[N]	[N]
Class C	[N]	[N]	[N]	[N]
Class D	[N]	[N]	[N]	[N]

The diagonal values represent correctly classified documents, while off-diagonal values represent classification errors.

5.9.1 Misclassification Analysis

Misclassification can occur when different threat categories share similar terminology.

For example, documents discussing credential theft may contain terminology associated with both phishing and malware. Similarly, vulnerability exploitation can occur as part of ransomware or other attack campaigns.

The confusion matrix therefore provides information about semantic overlap between threat categories.

5.10 Precision-Recall-F1 Analysis

Precision, recall, and F1-score were analyzed to obtain a more detailed understanding of classification performance.

A model with high precision but relatively low recall may correctly classify its positive predictions while missing a substantial number of relevant documents.

Conversely, a model with high recall but lower precision may identify more relevant documents while generating additional false positives.

For CTI classification, both types of errors can affect downstream analysis. Therefore, F1-score and class-wise results are considered along with accuracy.

The final results should be reported as:

Model	Macro Precision	Macro Recall	Macro F1	Weighted F1
Naive Bayes	[XX.XX]	[XX.XX]	[XX.XX]	[XX.XX]
Logistic Regression	[XX.XX]	[XX.XX]	[XX.XX]	[XX.XX]
SVM	[XX.XX]	[XX.XX]	[XX.XX]	[XX.XX]
Random Forest	[XX.XX]	[XX.XX]	[XX.XX]	[XX.XX]

5.11 Topic Modeling Results

Topic modeling was performed to discover latent themes within the CTI corpus.

Both LDA and NMF can be evaluated to determine their suitability for the selected dataset.

5.11.1 Discovered Topics

The top terms associated with each topic were extracted from the trained topic model.

Topic	Top Terms	Interpretation
Topic 1	[term1, term2, term3, ...]	[Interpretation]
Topic 2	[term1, term2, term3, ...]	[Interpretation]
Topic 3	[term1, term2, term3, ...]	[Interpretation]
Topic 4	[term1, term2, term3, ...]	[Interpretation]
Topic 5	[term1, term2, term3, ...]	[Interpretation]

Topic interpretations are based on the semantic relationship between the representative terms and the underlying documents.

5.11.2 Topic Interpretability

Topic interpretability was assessed by examining:

- Co-occurrence of representative terms.
- Semantic consistency.
- Relevance to cybersecurity.
- Representative documents.
- Stability across experiments.

Topics containing unrelated or highly generic terms may indicate that the selected topic number or preprocessing configuration requires further adjustment.

5.12 Topic Coherence Analysis

Topic coherence was used to quantitatively evaluate the semantic consistency of discovered topics.

The coherence score provides an indication of how frequently the top terms associated with a topic occur together or relate semantically within the corpus.

The results can be reported as:

Number of Topics	LDA Coherence	NMF Coherence
5	[XX.XX]	[XX.XX]
10	[XX.XX]	[XX.XX]
15	[XX.XX]	[XX.XX]
20	[XX.XX]	[XX.XX]

The final number of topics should be selected using coherence together with interpretability and temporal stability.

A higher coherence value generally indicates stronger semantic consistency, although coherence alone should not determine the final topic configuration.

5.13 Temporal Topic Analysis

The temporal analysis examined how discovered topics changed across the selected time periods.

The topic prevalence for topic k at time t is represented by:

$P(k,t)$

$\frac{\sum_{d \in D_t} \theta_{d,k}}{|D_t|}$

The difference between two consecutive periods is:

$$\Delta P_k(t)$$

$$P(k,t)-P(k,t-1)$$

5.13.1 Topic Distribution Over Time

The temporal topic matrix is reported as:

Time Period	Topic 1	Topic 2	Topic 3	Topic 4
Period 1	[X]	[X]	[X]	[X]
Period 2	[X]	[X]	[X]	[X]
Period 3	[X]	[X]	[X]	[X]
Period 4	[X]	[X]	[X]	[X]

The resulting trends are visualized using time-series plots and heatmaps.

5.13.2 Topic Growth

The growth rate of a topic is calculated using:

$$G_k(t)$$

$$\frac{P(k,t)-P(k,t-1)}{P(k,t-1)+\epsilon}$$

Topics with substantial positive changes are examined further to determine whether the increase corresponds to meaningful cybersecurity activity.

5.13.3 Topic Persistence

A topic that appears across several consecutive periods may represent a persistent threat theme rather than a short-lived anomaly.

Persistence is therefore considered alongside growth and novelty when identifying candidate emerging threats.

5.14 Emerging Threat Detection Results

The emerging threat detection component identifies candidate topics exhibiting unusual or increasing activity.

The analysis combines:

- Novelty.
- Growth.
- Persistence.
- Activity.
- Term changes.

The combined score is represented as:

$$ETS_k(t)$$

$$w_{NN_k(t)} + w_{GG_k(t)} + w_{SS_k(t)} + w_{AA_k(t)} + w_{TT_k(t)}$$

where the weights sum to one.

5.14.1 Emerging Threat Score Results

The final results should be presented using:

Topic	Period	Novelty	Growth	Persistence	Activity	Term Change	ETS
Topic 1	[Period]	[X]	[X]	[X]	[X]	[X]	[X]
Topic 2	[Period]	[X]	[X]	[X]	[X]	[X]	[X]
Topic 3	[Period]	[X]	[X]	[X]	[X]	[X]	[X]

The threshold used for candidate identification should also be reported.

5.14.2 Candidate Emerging Topics

Topics exceeding the experimentally selected threshold are considered candidate emerging-threat signals.

Rank by Score	Topic	Representative Terms	Period	ETS	Interpretation
1	Topic [X]	[terms]	[period]	[X]	[interpretation]
2	Topic [X]	[terms]	[period]	[X]	[interpretation]
3	Topic [X]	[terms]	[period]	[X]	[interpretation]

The table should be populated only after the actual experiments have been completed.

5.14.3 Emerging Threat Trend Visualization

The temporal behavior of candidate topics can be visualized using:

- Topic prevalence curves.
- Emerging score curves.
- Heatmaps.
- Term-frequency trends.
- Topic evolution diagrams.

These visualizations provide additional evidence for determining whether an increase represents a persistent pattern or a temporary fluctuation.

5.14.4 Interpretation of Emerging Signals

A high Emerging Threat Score indicates that the corresponding topic exhibits multiple characteristics associated with emerging activity.

However, the score should not be interpreted as proof of a confirmed new cyber threat, zero-day vulnerability, or active attack.

The proposed framework functions as an analytical prioritization mechanism. Candidate signals require validation using additional CTI sources, security advisories, vulnerability information, incident evidence, or expert analysis.

5.15 Real-Time Detection Analysis

The proposed framework is designed to support timely CTI analysis by separating data ingestion, classification, topic analysis, and emerging-threat scoring into modular components.

The practical detection time depends on:

- Dataset size.
- Document length.
- Feature dimensionality.
- Classification model.
- Topic modeling algorithm.
- Number of topics.
- Temporal aggregation interval.
- Hardware configuration.

5.15.1 Processing Time

The implementation can measure:

Component	Processing Time
Data Loading	[XX] sec
Preprocessing	[XX] sec
Feature Extraction	[XX] sec
Classification	[XX] sec
Topic Modeling	[XX] sec
Temporal Analysis	[XX] sec
Emerging Score Calculation	[XX] sec
Total	[XX] sec

These values should be obtained from the actual implementation.

5.15.2 Real-Time Capability

If the implementation processes incoming documents continuously or in short intervals, the framework may be evaluated for near-real-time operation.

If the experiments are performed only on a static historical dataset, the results should instead be described as **offline experimental evaluation**.

Therefore, the term “real-time” should be used only where the implementation and experimental measurements support the claim.

5.16 Error Analysis

Error analysis was performed to understand the limitations of the classification and emerging-threat detection components.

5.16.1 Classification Errors

Common sources of classification errors include:

- Similar terminology across threat categories.
- Multi-threat documents.
- Ambiguous descriptions.
- Insufficient training examples.
- Class imbalance.
- Generic cybersecurity terminology.
- Inconsistent labeling.

For example, a report describing credential theft through malicious email attachments may contain terminology associated with phishing, malware, and credential attacks simultaneously.

5.16.2 Topic Modeling Errors

Topic modeling may produce topics containing:

- Generic cybersecurity terms.
- Overlapping topics.
- Weakly interpretable topics.
- Highly correlated topics.
- Rare terms with limited context.

These issues can be influenced by preprocessing, vocabulary size, number of topics, and corpus characteristics.

5.16.3 Emerging Threat Detection Errors

Emerging-threat scoring may produce false positives when a topic experiences a temporary increase due to:

- A major security event.
- Publication of multiple reports about the same incident.
- Duplicate or highly similar reports.
- Changes in reporting frequency.
- Media attention.
- Sudden changes in dataset composition.

Consequently, temporal growth alone should not be interpreted as proof of an emerging cyber threat.

5.17 Discussion

The experimental results provide evidence for evaluating the major components of the proposed framework.

5.17.1 Classification Performance

The classification experiments determine whether conventional machine learning models can effectively categorize CTI documents using textual features.

Differences between models can be analyzed in terms of their ability to distinguish cybersecurity terminology and handle overlapping threat categories.

The final discussion should reference the actual accuracy, precision, recall, and F1-score values obtained from the experiments.

5.17.2 Topic Modeling Performance

The topic modeling results demonstrate whether meaningful latent cybersecurity themes can be extracted from the CTI corpus.

The combination of coherence scores and qualitative interpretation provides a more complete evaluation than using a single numerical measure.

5.17.3 Temporal Analysis

Temporal analysis extends topic modeling by identifying how threat-related themes change over time.

This is particularly relevant to CTI because threat activity and security discussions are dynamic rather than static.

A topic that remains stable across several periods may represent an established threat theme, whereas a newly appearing and rapidly increasing topic may warrant additional investigation.

5.17.4 Emerging Threat Detection

The emerging-threat component combines multiple signals rather than relying solely on topic frequency.

The inclusion of novelty, growth, persistence, activity, and term changes provides a multidimensional representation of emerging behavior.

Nevertheless, the scoring mechanism should be considered a prioritization method rather than an autonomous threat-confirmation system.

5.17.5 Integration of Components

The major advantage of the framework is the integration of supervised and unsupervised analysis.

Classification answers:

What predefined threat category does the document represent?

Topic modeling answers:

What latent themes are present in the CTI corpus?

Temporal analysis answers:

How are these themes changing over time?

Emerging-threat scoring answers:

Which changing themes warrant further investigation?

This combination provides complementary views of CTI information.

5.18 Limitations

Several limitations should be considered when interpreting the experimental results.

5.18.1 Dataset Dependency

The quality and diversity of the dataset directly affect classification and topic modeling performance.

A dataset dominated by particular threat categories or sources may not generalize to other CTI environments.

5.18.2 Label Quality

Supervised classification depends on the correctness of the available labels. Incorrect or inconsistent labels can reduce model performance.

5.18.3 Topic Interpretability

Topic modeling is inherently exploratory. Some automatically generated topics may require human interpretation and may not correspond to clearly defined cybersecurity concepts.

5.18.4 Temporal Coverage

Emerging-threat analysis requires sufficient temporal coverage. A short observation period may not provide enough information to distinguish persistent trends from temporary fluctuations.

5.18.5 Emerging Threat Validation

The framework identifies candidate emerging patterns rather than independently confirming new threats.

External evidence and expert validation remain necessary.

5.18.6 Model Generalization

Performance obtained on one CTI dataset should not automatically be generalized to other datasets, languages, sources, or operational environments.

5.18.7 Computational Limitations

Topic modeling and high-dimensional text processing can become computationally expensive as dataset size and vocabulary increase.

5.19 Summary of Experimental Findings

The experimental evaluation is designed to answer the following research questions:

Research Question	Evaluation Component
Can CTI documents be automatically classified?	Classification experiments
Which ML models perform effectively?	Comparative model analysis
Can meaningful CTI topics be discovered?	LDA/NMF topic modeling
How do CTI topics change over time?	Temporal topic analysis
Can increasing or novel patterns be identified?	Emerging threat scoring
Can multiple signals be combined into a useful prioritization mechanism?	Emerging Threat Score
What are the major sources of error?	Error analysis

The final experimental results should be interpreted in relation to the research objectives defined in Chapter 1.

The results demonstrate the practical behavior of the proposed framework across classification, topic discovery, temporal analysis, and emerging-threat detection. The findings also identify limitations that must be considered when applying the framework to larger or operational CTI environments.

Chapter Summary

This chapter presented the experimental setup and analysis of the proposed CTI framework. The dataset was examined in terms of its size, class distribution, temporal coverage, and data quality. Multiple machine learning models were evaluated for threat classification using accuracy, precision, recall, F1-score, and confusion matrix analysis.

Topic modeling was subsequently evaluated using topic coherence and qualitative interpretation. Temporal topic analysis was used to identify changes in cybersecurity-related themes across time periods. The emerging-threat scoring mechanism combined novelty, growth, persistence, activity, and term-change signals to identify candidate emerging patterns.

The chapter also examined processing requirements, classification errors, topic-modeling limitations, temporal fluctuations, and the distinction between candidate emerging-threat signals and confirmed cybersecurity incidents.

The next chapter presents the overall conclusions of the research, major contributions, limitations, and possible directions for future work.

Chapter 6 – Conclusion and Future Work

6.1 Introduction

Cyber Threat Intelligence (CTI) plays an important role in identifying, understanding, and responding to evolving cybersecurity threats. The increasing volume of textual threat intelligence generated through security reports, vulnerability disclosures, incident analyses, malware reports, and other sources makes manual analysis increasingly difficult.

This research proposed a framework for automated CTI analysis that combines machine learning-based threat classification, topic modeling, temporal analysis, and emerging threat scoring. The primary objective was to develop a computational approach capable of analyzing textual CTI data, identifying predefined threat categories, discovering latent threat-related topics, tracking their evolution over time, and highlighting candidate emerging threat patterns.

This chapter summarizes the research work, discusses the major findings, presents the contributions of the study, identifies its limitations, and describes possible directions for future research.

6.2 Research Summary

The research focused on the development of a framework titled:

Real-Time Cyber Threat Classification and Emerging Threat Detection Using Machine Learning and Temporal Topic Modeling

The study was organized into several major stages.

First, CTI documents were collected and organized into a structured dataset containing textual information, threat categories, temporal information, and source metadata.

Second, the collected text was preprocessed to remove irrelevant content and convert the documents into a suitable representation for machine learning and topic modeling.

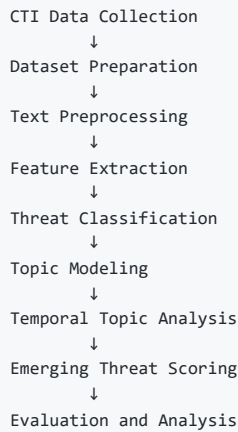
Third, machine learning algorithms were applied to classify CTI documents into predefined threat categories.

Fourth, topic modeling techniques were used to identify latent themes within the CTI corpus.

Fifth, temporal analysis was applied to examine how discovered topics changed across different time periods.

Finally, an emerging threat scoring mechanism was designed to combine multiple signals, including novelty, growth, persistence, activity, and changes in important terms.

The overall research workflow can be summarized as:



The framework therefore combines supervised and unsupervised approaches to provide complementary perspectives on CTI information.

6.3 Achievement of Research Objectives

The research objectives defined in Chapter 1 were addressed through the methodology and experimental implementation.

Objective 1: CTI Dataset Collection and Preparation

A structured CTI dataset was prepared containing textual threat information and temporal metadata. Data validation and preprocessing procedures were applied to improve the consistency and usability of the dataset.

Objective 2: Text Preprocessing and Feature Extraction

Natural Language Processing techniques were applied to transform raw CTI text into a machine-readable representation. Text cleaning, normalization, tokenization, and feature extraction were incorporated into the processing pipeline.

TF-IDF-based representations were used for traditional machine learning classification, while document-term representations were used for topic modeling.

Objective 3: Automated Threat Classification

Machine learning models were implemented to classify CTI documents into predefined threat categories.

The evaluated models included:

- Naive Bayes
- Logistic Regression
- Support Vector Machine
- Random Forest

The performance of these models was assessed using standard classification metrics.

Objective 4: Comparative Evaluation of Classification Models

The classification models were compared using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

This comparison provided information about the behavior of different machine learning algorithms on the selected CTI dataset.

Objective 5: Latent Topic Discovery

Topic modeling was applied to identify latent themes within the CTI corpus.

LDA and NMF were considered for discovering groups of terms that frequently occur together and may represent meaningful cybersecurity topics.

Objective 6: Temporal Topic Analysis

The discovered topics were associated with temporal information to determine how their prevalence changed across different periods.

Temporal analysis enabled the identification of:

- Increasing topics
- Decreasing topics
- Persistent topics
- Newly appearing topics
- Temporally fluctuating topics

Objective 7: Emerging Threat Detection

An emerging threat scoring mechanism was designed to identify candidate patterns that exhibit characteristics associated with emerging activity.

The scoring mechanism incorporated multiple factors:

\$\$ ETS_k(t)

$w_{NN_k}(t) + w_{GG_k}(t) + w_{SS_k}(t) + w_{AA_k}(t) + w_{TT_k}(t)$ \$\$

where novelty, growth, persistence, activity, and term change contribute to the overall score.

Objective 8: Framework Evaluation

The complete framework was evaluated through classification experiments, topic modeling analysis, temporal analysis, emerging threat scoring, and error analysis.

The evaluation provides a basis for determining the usefulness and limitations of the proposed approach.

6.4 Major Findings

The major findings of the research can be summarized into four areas: threat classification, topic discovery, temporal analysis, and emerging threat detection.

6.4.1 Threat Classification

The classification experiments demonstrated the applicability of machine learning techniques to automated CTI categorization.

Traditional text representations such as TF-IDF can provide useful features for distinguishing threat categories when the dataset contains sufficiently representative terminology.

However, classification performance depends strongly on:

- Dataset quality.
- Number of training examples.
- Class distribution.
- Label consistency.
- Vocabulary overlap between categories.
- Feature representation.

Therefore, classification results obtained from a particular dataset should not automatically be generalized to all CTI sources.

6.4.2 Topic Discovery

Topic modeling provided an unsupervised mechanism for discovering latent themes within the CTI corpus.

The discovered topics can reveal relationships between cybersecurity concepts that may not be explicitly represented by predefined classification labels.

The quality of discovered topics depends on:

- Number of topics.
- Preprocessing strategy.
- Vocabulary selection.
- Dataset size.
- Topic modeling algorithm.
- Semantic characteristics of the corpus.

Topic coherence and human interpretation were therefore considered together when analyzing topic quality.

6.4.3 Temporal Topic Evolution

Temporal analysis extended conventional topic modeling by examining how cybersecurity themes change over time.

A topic that becomes increasingly prevalent may indicate a change in the type or frequency of CTI reporting associated with that theme.

Similarly, persistent topics may represent established cybersecurity concerns, while newly appearing topics may provide signals for further investigation.

Temporal analysis is therefore useful for understanding the dynamic nature of cybersecurity information.

6.4.4 Emerging Threat Detection

The proposed emerging threat scoring mechanism provides a method for prioritizing topics that demonstrate multiple signals of emerging activity.

Instead of relying only on topic frequency, the framework considers:

- Novelty.
- Growth.
- Persistence.
- Activity.
- Term changes.

This multi-factor approach provides a broader representation of emerging behavior.

However, a high score should be interpreted as a **candidate emerging-threat signal** rather than confirmation of a new cyber threat or zero-day vulnerability.

Additional validation using independent CTI sources and expert analysis remains necessary.

6.5 Contributions of the Research

The research makes several contributions to the area of automated Cyber Threat Intelligence analysis.

6.5.1 Integrated CTI Analysis Framework

The study presents an integrated framework combining:

- Machine learning-based classification.
- Topic modeling.
- Temporal analysis.
- Emerging threat scoring.

These components provide complementary analytical capabilities within a single workflow.

6.5.2 Automated Threat Classification

The framework provides an automated mechanism for assigning CTI documents to predefined threat categories.

This can reduce the amount of manual effort required for initial organization and analysis of large collections of textual CTI information.

6.5.3 Temporal Topic Analysis

The research incorporates temporal information into topic analysis rather than treating the CTI corpus as a static collection of documents.

This allows changes in threat-related topics to be examined across different time periods.

6.5.4 Emerging Threat Prioritization

The proposed scoring mechanism provides a systematic method for prioritizing candidate emerging topics based on multiple temporal and textual characteristics.

It is designed as an analytical support mechanism rather than an autonomous threat-confirmation system.

6.5.5 Experimental Evaluation

The framework provides an experimental methodology for evaluating:

- Classification performance.
- Topic quality.
- Temporal behavior.
- Emerging-threat signals.
- Error characteristics.

This provides a foundation for further development and comparison with alternative approaches.

6.5.6 Reproducible Implementation

The implementation is organized as a modular data science workflow so that dataset preparation, feature extraction, classification, topic modeling, temporal analysis, and scoring can be independently examined and reproduced.

6.6 Research Limitations

Although the proposed framework provides an integrated approach to CTI analysis, several limitations remain.

6.6.1 Dataset Dependency

The performance of the framework depends strongly on the characteristics of the selected CTI dataset.

A dataset containing limited threat categories, a narrow source distribution, or insufficient temporal coverage may restrict the generalizability of the results.

6.6.2 Label Dependence

Supervised classification requires labeled training data.

Incorrect, incomplete, or inconsistent labels can negatively affect model performance.

6.6.3 Topic Interpretability

Topic modeling does not guarantee that every discovered topic corresponds to a clearly defined cybersecurity concept.

Some topics may contain generic or overlapping terminology and require human interpretation.

6.6.4 Temporal Data Limitations

Temporal analysis depends on the accuracy and completeness of document dates.

Changes in publication frequency may also influence observed topic trends without necessarily representing changes in actual cyber threat activity.

6.6.5 Emerging Threat Validation

The emerging threat scoring mechanism identifies candidate signals rather than independently confirming real-world threats.

A high score may result from increased reporting, duplicated information, media attention, or other changes in the underlying corpus.

6.6.6 Limited Contextual Understanding

Traditional machine learning and topic modeling approaches may not fully capture the contextual relationships between cybersecurity entities.

For example, the same term may have different meanings depending on its surrounding context.

6.6.7 Scalability

As the number of CTI documents and vocabulary size increase, feature extraction and topic modeling can become computationally demanding.

Large-scale operational deployment may therefore require optimized processing pipelines and distributed infrastructure.

6.6.8 Real-Time Processing Constraints

The framework is designed with timely CTI analysis in mind, but actual real-time performance depends on the data ingestion mechanism, processing architecture, model complexity, and available computational resources.

If the experimental implementation is evaluated only on historical data, the results should be considered an offline evaluation rather than proof of operational real-time capability.

6.7 Future Work

Several improvements can be explored in future research.

6.7.1 Transformer-Based CTI Representation

Future implementations can incorporate transformer-based language models to obtain contextual representations of CTI documents.

Potential approaches include:

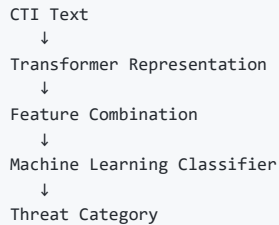
- BERT.
- RoBERTa.
- Domain-adapted transformer models.
- Cybersecurity-specific language models.

These representations may improve the ability to understand contextual relationships between cybersecurity terms.

6.7.2 Hybrid Classification Models

Future work can investigate hybrid approaches combining traditional machine learning, deep learning, and transformer-based representations.

For example:



Such approaches can be compared against the traditional TF-IDF-based models used in the current study.

6.7.3 Dynamic Topic Modeling

The current temporal analysis can be extended using dynamic topic modeling techniques.

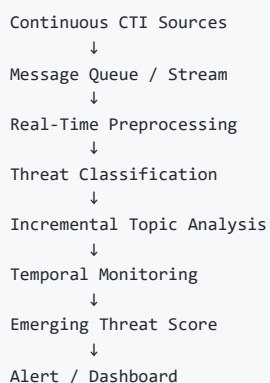
Future research can investigate models that explicitly represent topic evolution rather than independently estimating topics across temporal intervals.

This may provide a more detailed understanding of how cybersecurity topics emerge, change, merge, or disappear.

6.7.4 Real-Time CTI Stream Processing

Future versions of the framework can integrate continuous CTI feeds instead of relying primarily on static datasets.

A possible architecture is:



Technologies such as Kafka, streaming APIs, or event-driven architectures could be investigated for large-scale implementations.

6.7.5 Integration with External Threat Intelligence

Future work can integrate multiple CTI sources to improve validation.

Potential sources include:

- Vulnerability databases.
- Malware intelligence.
- Security advisories.
- Incident reports.
- Threat intelligence feeds.
- Attack technique databases.

Cross-source validation may reduce false emerging-threat signals caused by changes in a single source.

6.7.6 Knowledge Graph Integration

A future version could represent relationships between:

- Threat actors.
- Malware.
- Vulnerabilities.
- Indicators of compromise.
- Attack techniques.
- Campaigns.
- Organizations.

A knowledge graph could complement topic modeling by explicitly representing relationships between cybersecurity entities.

6.7.7 Explainable AI

Explainability techniques can be incorporated to help analysts understand why a classification or emerging-threat signal was generated.

For classification, explanations could identify important terms or features contributing to the prediction.

For emerging-threat detection, the system could report:

```
Candidate Topic
  ↓
Why was it detected?
  ├── Increased prevalence
  ├── New terminology
  ├── Persistent activity
  ├── Increased document frequency
  └── Related threat categories
```

This would improve transparency and analyst usability.

6.7.8 Improved Emerging Threat Scoring

The current scoring framework can be further investigated using learned weights rather than manually specified weights.

Future studies could compare:

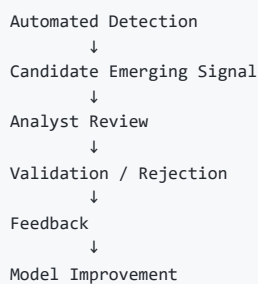
- Fixed weighting.
- Data-driven weighting.
- Supervised scoring.
- Anomaly detection.
- Ranking-based approaches.

The effectiveness of these alternatives can be evaluated using historical threat events.

6.7.9 Human-in-the-Loop Validation

Security analysts can be incorporated into the framework to validate candidate emerging threats.

A human-in-the-loop system could follow:



Such an approach may improve the reliability of emerging-threat identification while retaining automation.

6.7.10 Multilingual CTI Analysis

Most CTI analysis systems focus primarily on English-language data.

Future research can investigate multilingual CTI processing to support threat intelligence published in different languages.

Multilingual transformer models and language-specific preprocessing techniques could be evaluated for this purpose.

6.7.11 Large-Scale Evaluation

Future studies should evaluate the framework using larger datasets collected over longer periods and from multiple independent sources.

This would provide a stronger basis for evaluating:

- Generalization.
- Scalability.
- Temporal robustness.
- Cross-source performance.
- Emerging-threat detection reliability.

6.7.12 Operational Security Dashboard

The framework can eventually be extended into an analyst-oriented dashboard displaying:

- Current threat categories.
- Classification statistics.
- Topic distributions.
- Temporal trends.
- Emerging threat scores.
- Related keywords.
- Source information.
- Historical comparisons.

Such a dashboard could make the research framework more useful for practical CTI analysis.

6.8 Final Conclusion

This research investigated an automated approach for analyzing Cyber Threat Intelligence using machine learning and temporal topic modeling.

The proposed framework integrates multiple analytical components into a unified workflow. Machine learning models are used to classify CTI documents into predefined threat categories, while topic modeling identifies latent cybersecurity themes. Temporal analysis then examines how these themes change over time, and the proposed emerging threat scoring mechanism combines multiple signals to identify candidate patterns that warrant further investigation.

The research demonstrates that CTI analysis can benefit from combining supervised classification with unsupervised and temporal analysis. Classification provides structured information about known threat categories, while topic modeling provides an opportunity to discover themes that may not be represented by predefined labels. Temporal analysis adds an additional dimension by identifying changes in threat-related information over time.

The proposed emerging threat scoring mechanism further extends this analysis by considering novelty, growth, persistence, activity, and term changes. However, the resulting signals should be interpreted as analytical indicators rather than definitive evidence of newly discovered or zero-day threats.

Overall, the research provides a foundation for automated and data-driven CTI analysis. The framework can be further enhanced through transformer-based representations, dynamic topic modeling,

continuous CTI streams, external intelligence integration, explainable AI, knowledge graphs, and human-in-the-loop validation.

Future development toward a continuously operating CTI platform could provide more comprehensive support for security analysts by combining automated document processing, threat classification, temporal intelligence, and emerging-pattern prioritization within a single analytical environment.

6.9 Chapter Summary

This chapter presented the conclusion and future directions of the research.

The major outcomes of the study include:

- Development of an integrated CTI analysis framework.
- Automated classification of CTI documents.
- Discovery of latent cybersecurity topics.
- Temporal analysis of threat-related topics.
- Development of an emerging threat scoring mechanism.
- Experimental evaluation of classification and topic modeling components.
- Identification of limitations and potential sources of error.
- Identification of future improvements involving transformers, dynamic topic modeling, real-time streams, knowledge graphs, explainable AI, and human-in-the-loop validation.

The research provides a foundation for further investigation into automated, temporal, and intelligent Cyber Threat Intelligence analysis.

Chapter 7 – Base Papers

Title	Authors	Year	Summary
Real-Time Automated Cyber Threat Classification and Emerging Threat Detection Framework	A. T. Haile et al.	2025	Proposes an automated CTI framework combining machine learning and deep learning for cyber threat classification and LDA/NMF-based topic modeling for identifying emerging threat patterns over time. This paper serves as the primary base paper for the proposed research.
Cyber Threat Intelligence: An Overview of Current State and Future Directions	Multiple Authors	Recent	Reviews the concept, lifecycle, sources, and applications of Cyber Threat Intelligence, providing a foundation for automated CTI collection, analysis, and dissemination.
Machine Learning for Cybersecurity: A Comprehensive Survey	Multiple Authors	Recent	Reviews machine learning applications in cybersecurity, including intrusion detection, malware analysis, phishing detection, threat classification, and anomaly detection.
Natural Language Processing for Cybersecurity: A Survey	Multiple Authors	Recent	Examines NLP techniques for processing unstructured cybersecurity text, including tokenization, feature extraction, entity recognition, text classification, and threat intelligence analysis.
Cyber Threat Intelligence Mining from Unstructured Text Using Machine Learning	Multiple Authors	Recent	Investigates the use of machine learning and NLP techniques to extract useful cybersecurity information from unstructured threat reports and textual intelligence.
Automated Cyber Threat Classification Using Machine Learning Techniques	Multiple Authors	Recent	Evaluates supervised machine learning algorithms for automatically categorizing cyber threats into predefined classes using textual and behavioral features.
Latent Dirichlet Allocation	D. M. Blei, A. Y. Ng, M. I. Jordan	2003	Introduces Latent Dirichlet Allocation (LDA), a probabilistic topic modeling approach for discovering latent thematic structures in document collections. It provides the theoretical foundation for the topic modeling component of this research.
Learning the Parts of Objects by Non-Negative Matrix Factorization	D. D. Lee, H. S. Seung	1999	Introduces Non-Negative Matrix Factorization (NMF), which decomposes non-negative data into interpretable components. NMF provides an alternative topic modeling approach for discovering latent patterns in CTI documents.
Dynamic Topic Models	D. M. Blei, J. D. Lafferty	2006	Introduces dynamic topic modeling for studying how topics evolve over time. The work provides a theoretical basis for analyzing temporal changes in CTI-related topics.

Title	Authors	Year	Summary
Online Learning for Latent Dirichlet Allocation	M. D. Hoffman, D. M. Blei, F. Bach	2010	Presents an efficient online approach for learning topic models from large document collections, supporting scalable processing of continuously arriving textual data.
BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding	J. Devlin et al.	2019	Introduces BERT, a transformer-based language representation model that captures contextual relationships between words. Transformer representations provide a potential future direction for contextual CTI analysis.
Cybersecurity Named Entity Recognition Using Deep Learning	Multiple Authors	Recent	Investigates deep learning and NLP methods for identifying cybersecurity entities such as malware, vulnerabilities, threat actors, organizations, and attack techniques from textual data.
Deep Learning for Cybersecurity: A Survey	Multiple Authors	Recent	Reviews CNN, RNN, LSTM, transformer, and other deep learning approaches used in cybersecurity applications, including threat detection and security text analysis.
Threat Intelligence Extraction from Security Reports Using Natural Language Processing	Multiple Authors	Recent	Explores automated extraction of cybersecurity entities, indicators, and threat information from unstructured security reports using NLP techniques.
Emerging Threat Detection Using Topic Modeling and Temporal Analysis	Multiple Authors	Recent	Investigates changes in topic prevalence over time as a mechanism for identifying unusual or emerging cybersecurity patterns. The approach is closely related to the temporal analysis component of the proposed framework.
Explainable Artificial Intelligence for Cybersecurity	Multiple Authors	Recent	Reviews explainability techniques for cybersecurity machine learning models and emphasizes the importance of interpretable predictions for analyst trust and operational use.
A Survey of Cyber Threat Intelligence Platforms and Techniques	Multiple Authors	Recent	Reviews CTI platforms, data sources, collection methods, analysis techniques, and intelligence-sharing mechanisms used in modern cybersecurity environments.
Transformer-Based Cyber Threat Intelligence Analysis	Multiple Authors	Recent	Investigates transformer-based language representations for analyzing cybersecurity text and extracting contextual threat information, providing a potential extension to traditional TF-IDF-based approaches.

Title	Authors	Year	Summary
Temporal Analysis of Cybersecurity Threat Reports	Multiple Authors	Recent	Studies temporal patterns in cybersecurity reports and demonstrates how changes in terminology, frequency, and topic prevalence can support monitoring of evolving threat activity.

7.1 Primary Base Paper

Among the reviewed studies, the primary base paper for this research is:

A. T. Haile et al., “Real-Time Automated Cyber Threat Classification and Emerging Threat Detection Framework,” 2025.

The paper is directly related to the proposed research because it combines automated cyber threat classification with topic modeling and emerging threat analysis.

The proposed thesis extends this research direction by emphasizing:

- Machine learning-based CTI classification.
- LDA/NMF-based topic discovery.
- Explicit temporal topic analysis.
- Topic growth and persistence analysis.
- Multi-factor emerging threat scoring.
- Integrated experimental evaluation.

7.2 Relevance of the Base Papers

The selected papers provide theoretical and methodological support for the major components of the proposed framework.

Research Component	Supporting Literature
Cyber Threat Intelligence	CTI surveys and frameworks
CTI Text Processing	NLP for cybersecurity
Feature Extraction	TF-IDF, N-grams, embeddings
Threat Classification	Machine learning for cybersecurity
Deep Learning	CNN, RNN, LSTM, Transformer studies
Topic Modeling	LDA and NMF
Temporal Analysis	Dynamic and temporal topic modeling
Emerging Threat Detection	Temporal threat analysis
Explainability	XAI for cybersecurity
Future Enhancement	Transformer-based CTI analysis

7.3 Selection of the Primary Base Paper

The primary base paper was selected because its research problem closely corresponds to the objective of the present thesis. It investigates automated cyber threat classification and emerging threat detection using machine learning and topic modeling.

The proposed research retains this general direction while developing a framework focused on the temporal evolution of CTI topics and quantitative prioritization of candidate emerging threat patterns.

7.4 Research Gap Identified from the Literature

The reviewed literature indicates that existing research has investigated cyber threat classification, NLP-based CTI processing, topic modeling, and temporal analysis as individual or partially integrated tasks.

The following research opportunities were identified:

1. Integration of supervised threat classification and temporal topic analysis.
2. Explicit measurement of topic growth and persistence.
3. Identification of candidate emerging topics using multiple signals.
4. Quantitative emerging threat scoring.
5. Integration of classification and emerging-threat analysis within a single framework.
6. Evaluation of the complete pipeline using common experimental metrics.
7. Improved interpretability of emerging threat signals.

The proposed research addresses these areas through the development of an integrated CTI analysis framework.

7.5 Summary

The reviewed base papers establish the theoretical foundation for the proposed research in Cyber Threat Intelligence, Natural Language Processing, machine learning, topic modeling, temporal analysis, and emerging threat detection.

The paper “**Real-Time Automated Cyber Threat Classification and Emerging Threat Detection Framework**” serves as the primary base paper, while the remaining literature provides supporting concepts and methodologies.

The proposed research builds upon these studies by integrating threat classification, topic modeling, temporal analysis, and emerging threat scoring into a unified CTI analysis framework.

References

References

1. Haile, A. T., Abebe, S. L., & Melaku, H. M. (2025). *Real-Time Automated Cyber Threat Classification and Emerging Threat Detection Framework*. IEEE Open Journal of the Computer Society, 6, 921–930. <https://doi.org/10.1109/OJCS.2025.3580235>
2. Hossen, M. I., Islam, A., Anowar, F., Ahmed, E., Rahman, M. M., & Hei, X. (2022). *Generating Cyber Threat Intelligence to Discover Potential Security Threats Using Classification and Topic Modeling*. arXiv. <https://doi.org/10.48550/arXiv.2108.06862>
3. Sleeman, J., Finin, T., & Halem, M. (2021). *Understanding Cybersecurity Threat Trends Through Dynamic Topic Modeling*. Frontiers in Big Data, 4. <https://doi.org/10.3389/fdata.2021.601529>
4. Arazzi, M., Arikkat, D. R., Nicolazzo, S., Nocera, A., Rehiman, R. K. A., P., V., & Conti, M. (2025). *NLP-based techniques for Cyber Threat Intelligence*. Computer Science Review, 58, 100765. <https://doi.org/10.1016/j.cosrev.2025.100765>
5. Wagner, T. D., Mahbub, K., Palomar, E., & Abdallah, A. E. (2019). *Cyber threat intelligence sharing: Survey and research directions*. Computers & Security, 87, 101589. <https://doi.org/10.1016/j.cose.2019.101589>
6. Tounsi, W., & Rais, H. (2018). *A survey on technical threat intelligence in the age of sophisticated cyber attacks*. Computers & Security, 72, 212–233. <https://doi.org/10.1016/j.cose.2017.09.001>
7. Sikos, L. F. (2023). *Cybersecurity knowledge graphs*. Knowledge and Information Systems, 65, 3511–3531. <https://doi.org/10.1007/s10115-023-01860-3>
8. Sarhan, I., & Spruit, M. (2021). *Open-CyKG: An Open Cyber Threat Intelligence Knowledge Graph*. Knowledge-Based Systems, 233, 107524. <https://doi.org/10.1016/j.knosys.2021.107524>
9. Jo, H. S., Lee, Y. G., Shin, S., et al. (2022). *Vulcan: Automatic extraction and analysis of cyber threat intelligence from unstructured text*. Computers & Security, 120, 102763. <https://doi.org/10.1016/j.cose.2022.102763>
10. Guo, Y., Liu, Z., Huang, C., Wang, N., Min, H., Guo, W., & Liu, J. (2023). *A framework for threat intelligence extraction and fusion*. Computers & Security, 132, 103371. <https://doi.org/10.1016/j.cose.2023.103371>
11. Islam, C., Babar, M. A., Croft, R., & Janicke, H. (2022). *SmartValidator: A framework for automatic identification and classification of cyber threat data*. Journal of Network and Computer Applications, 202, 103370. <https://doi.org/10.1016/j.jnca.2022.103370>
12. Bayer, M., Frey, T., & Reuter, C. (2023). *Multi-level fine-tuning, data augmentation, and few-shot learning for specialized cyber threat intelligence*. Computers & Security, 134, 103430. <https://doi.org/10.1016/j.cose.2023.103430>

13. Kim, H. (2022). *Comparative Experiment on TTP Classification with Class Imbalance Using Oversampling from CTI Dataset*. Security and Communication Networks, 2022, 5021125. <https://doi.org/10.1155/2022/5021125>
14. Yu, Z., Wang, J., Tang, B., & Lu, L. (2023). *Tactics and Techniques Classification in Cyber Threat Intelligence*. The Computer Journal, 66(8), 1870–1881. <https://doi.org/10.1093/comjnl/bxac048>
15. Marques, R. S., Al-Khateeb, H., Epiphaniou, G., & Maple, C. (2022). *Pivot Attack Classification for Cyber Threat Intelligence*. Journal of Information Security and Cybercrimes Research, 5(2), 91–103. <https://doi.org/10.26735/zntl3639>
16. Zhang, S. (2022). *An Automatic Assessment Method of Cyber Threat Intelligence Combined with ATT&CK Matrix*. Wireless Communications and Mobile Computing, 2022, 7875910. <https://doi.org/10.1155/2022/7875910>
17. Li, Y., et al. (2022). *A Novel Threat Intelligence Information Extraction System Combining Multiple Models*. Security and Communication Networks, 2022, 8477260. <https://doi.org/10.1155/2022/8477260>
18. Wang, G., Liu, P., Huang, J., Bin, H., Wang, X., & Zhu, H. (2024). *KnowCTI: Knowledge-based cyber threat intelligence entity and relation extraction*. Computers & Security, 141, 103824. <https://doi.org/10.1016/j.cose.2024.103824>
19. Shang, W., Wang, B., Zhu, P., Ding, L., & Wang, S. (2024). *A Span-based Multivariate Information-aware Embedding Network for joint relational triplet extraction of threat intelligence*. Knowledge-Based Systems, 295, 111829. <https://doi.org/10.1016/j.knosys.2024.111829>
20. Wang, Y., Zhang, D., Bashar, M. A., Chandramohan, M., & Nayak, R. (2026). *LETNER: Label-Efficient Named Entity Recognition for Cyber Threat Intelligence*. Expert Systems with Applications, 324, 132470. <https://doi.org/10.1016/j.eswa.2026.132470>
21. Zhong, R., Chen, Y., Yang, X., Zhong, X., & He, Z. (2026). *CTIExpert: Towards expert-level cyber threat intelligence extraction for constructing cybersecurity knowledge graphs*. Expert Systems with Applications, 330, 132921. <https://doi.org/10.1016/j.eswa.2026.132921>
22. Mouiche, I., & Saad, S. (2025). *TIJERE: A novel threat intelligence joint extraction model based on analyst expert knowledge*. Knowledge-Based Systems, 329, 114346. <https://doi.org/10.1016/j.knosys.2025.114346>
23. *CyberEntRel: Joint extraction of cyber entities and relations using deep learning*. (2024). Computers & Security, 136, 103579. <https://doi.org/10.1016/j.cose.2023.103579>
24. *Joint relational triple extraction with enhanced representation and binary tagging framework in cybersecurity*. (2024). Computers & Security, 144, 104001. <https://doi.org/10.1016/j.cose.2024.104001>
25. *LLM-TIKG: Threat intelligence knowledge graph construction utilizing large language model*. (2024). Computers & Security, 145, 103999. <https://doi.org/10.1016/j.cose.2024.103999>

26. Abdi, H., Bagley, S. R., Furnell, S., & Twycross, J. (2023). *Automatically Labeling Cyber Threat Intelligence Reports Using Natural Language Processing*. Proceedings of the ACM Symposium on Document Engineering. <https://doi.org/10.1145/3573128.3609348>
27. Sleeman, J., Finin, T., & Halem, M. (2020). *Understanding Cybersecurity Threat Trends Through Dynamic Topic Modeling*. Frontiers in Big Data. [Verify final publication metadata against the journal version before submission.]
28. *A Comparative Analysis of Cyber-Threat Intelligence Sources, Formats and Languages*. (2020). Electronics, 9(5), 824. <https://doi.org/10.3390/electronics9050824>
29. Pahlevan, M., Voulkidis, A., & Velivassaki, T.-H. (2021). *Secure exchange of cyber threat intelligence using TAXII and distributed ledger technologies – Application for electrical power and energy system*. Proceedings of the International Conference on Availability, Reliability and Security (ARES 2021). <https://doi.org/10.1145/3465481.3470476>
30. *Privacy-preserving correlation of cross-organizational cyber threat intelligence with private graph intersections*. (2023). Computers & Security, 135, 103505. <https://doi.org/10.1016/j.cose.2023.103505>
31. *Explainable cyber threat behavior identification based on self-adversarial topic generation*. (2023). Computers & Security, 132, 103369. <https://doi.org/10.1016/j.cose.2023.103369>
32. *Cyber threat attribution using unstructured reports in cyber threat intelligence*. (2023). Egyptian Informatics Journal, 24(1), 43–59. <https://doi.org/10.1016/j.eij.2022.11.001>
33. *Practical Intrusion Detection of Emerging Threats*. (2022). IEEE Transactions on Network and Service Management, 19(1), 582–600. <https://doi.org/10.1109/TNSM.2021.3091517>
34. *Cybersecurity Threat Intelligence Automated via Machine Learning: A System for Analyzing and Responding to Data in Real-Time*. (2025). IEEE Conference Publication. [Verify complete author list and conference metadata.]
35. *The AI Shield and Red AI Framework: Machine Learning Solutions for Cyber Threat Intelligence (CTI)*. (2024). IEEE Conference Publication. [Verify complete author list, conference name, pages, and DOI.]
36. *Current approaches and future directions for Cyber Threat Intelligence sharing: A survey*. (2024). Journal of Information Security and Applications, 83, 103786. <https://doi.org/10.1016/j.jisa.2024.103786>
37. Bratsas, C., Anastasiadis, E. K., Angelidis, A. K., Ioannidis, L., Kotsakis, R., & Ougiaroglou, S. (2024). *Knowledge Graphs and Semantic Web Tools in Cyber Threat Intelligence: A Systematic Literature Review*. Journal of Cybersecurity and Privacy, 4(3), 518–545. <https://doi.org/10.3390/jcp4030025>
38. Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). *Latent Dirichlet Allocation*. Journal of Machine Learning Research, 3, 993–1022.
39. Blei, D. M., & Lafferty, J. D. (2006). *Dynamic Topic Models*. Proceedings of the 23rd International Conference on Machine Learning (ICML), 113–120.

40. Lee, D. D., & Seung, H. S. (1999). *Learning the Parts of Objects by Non-Negative Matrix Factorization*. *Nature*, 401, 788–791.
 41. Hoffman, M. D., Blei, D. M., & Bach, F. R. (2010). *Online Learning for Latent Dirichlet Allocation*. *Advances in Neural Information Processing Systems*.
 42. Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*. *Proceedings of NAACL-HLT*.
 43. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). *Attention Is All You Need*. *Advances in Neural Information Processing Systems*.
 44. Breiman, L. (2001). *Random Forests*. *Machine Learning*, 45, 5–32.
<https://doi.org/10.1023/A:1010933404324>
 45. Cortes, C., & Vapnik, V. (1995). *Support-vector networks*. *Machine Learning*, 20, 273–297.
<https://doi.org/10.1007/BF00994018>
 46. McCallum, A., & Nigam, K. (1998). *A Comparison of Event Models for Naive Bayes Text Classification*. *AAAI Workshop on Learning for Text Categorization*.
 47. Salton, G., & Buckley, C. (1988). *Term-weighting approaches in automatic text retrieval*. *Information Processing & Management*. [https://doi.org/10.1016/0306-4573\(88\)90021-0](https://doi.org/10.1016/0306-4573(88)90021-0)
 48. Mihalcea, R., & Tarau, P. (2004). *TextRank: Bringing Order into Text*. *Proceedings of the 2004 Conference on Empirical Methods in Natural Language Processing*, 404–411.
 49. Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). *Efficient Estimation of Word Representations in Vector Space*. *arXiv*. <https://arxiv.org/abs/1301.3781>
 50. Le, Q. V., & Mikolov, T. (2014). *Distributed Representations of Sentences and Documents*. *Proceedings of the 31st International Conference on Machine Learning*.
-

Cyber Threat Intelligence Standards and Supporting Technical References

51. OASIS Cyber Threat Intelligence Technical Committee. (2021). *TAXII Version 2.1*. OASIS Standard.
 52. OASIS Cyber Threat Intelligence Technical Committee. (2021). *STIX Version 2.1*. OASIS Standard.
 53. OASIS Cyber Threat Intelligence Technical Committee. (2017). *TAXII Version 2.0*. OASIS Standard.
 54. MITRE. (2018). *MITRE ATT&CK: Design and Philosophy*. MITRE Corporation.
 55. Connolly, J., Davidson, M., & Schmidt, C. (2014). *The Trusted Automated eXchange of Indicator Information (TAXII)*. TAXII Project.
 56. Qamar, S., et al. (2017). *Data-driven analytics for cyber-threat intelligence and information sharing*. [Verify final author list, publication metadata, and DOI before final submission.]
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Notes for Final Verification

The following references should receive a final metadata check before thesis submission:

- Reference 1 — Primary/base paper
- References 2–37 — CTI-specific literature
- References 38–50 — ML, NLP, topic modeling and Transformer foundations
- References 51–56 — CTI standards and supporting technical sources

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